

THE ONTARIO CURRICULUM

GRADES 9–12

Technological Education

This file is an extract and may not reflect or represent the full Ontario Curriculum.

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PDF versions of a curriculum include the following information from the [Curriculum and Resources website](#):

- the Program Planning and Assessment and Evaluation sections of the Curriculum and Resources website that apply to all Ontario curriculum, Grades 1–12;
- the Curriculum Context that is specific to a discipline;
- the strands of the curriculum; and
- glossaries and appendices as applicable.

The Ontario Curriculum Grades 9–12: Technological Education,

This curriculum policy supersedes *The Ontario Curriculum, Grades 9 and 10: Technological Education, 2009*. Beginning in the 2024–25 school year, all technological education courses for Grades 9 and 10 will be based on the expectations outlined in this curriculum. Technological education courses for Grades 11 and 12 will continue to be based on [The Ontario Curriculum, Grades 11 and 12: Technological Education, 2009](#). All references to Grades 9 and 10 that appear in that document have been superseded.

Version History:

Version Date	Description
May 2024	New courses issued for Grades 9 and 10 – Technology and the Skilled Trades, Open (TAS1O and TAS2O). Beginning in the 2024–25 school year, Exploring Technologies, Grade 9 (TIJ1O) should not be offered, and the associated course code will be expired. Beginning in the 2024–25 school year, all Grade 9 and 10 broad-based technology (BBT) focus courses will be delivered using the TAS1O/TAS2O expectations.

Program Planning and Assessment and Evaluation Content

Last updated: June 2023

This content is part of official issued curriculum providing the most up-to-date information (i.e., front matter). This content is applicable to all curriculum documents, Grades 1 to 12. Educators must consider this information to guide the implementation of curriculum and in creating the environment in which it is taught.

This curriculum policy supersedes *The Ontario Curriculum, Grades 9 and 10: Technological Education, 2009*. Beginning in the 2024–25 school year, all technological education courses for Grades 9 and 10 will be based on the expectations outlined in this curriculum. Technological education courses for Grades 11 and 12 will continue to be based on [*The Ontario Curriculum, Grades 11 and 12: Technological Education, 2009*](#). All references to Grades 9 and 10 that appear in that document have been superseded.

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Une publication équivalente est disponible en français sous le titre suivant : Le curriculum de l'Ontario, 9^e et 10^e année – Éducation technologique, 2024 et Le curriculum de l'Ontario, 11^e et 12^e année – Éducation technologique, 2009.

Considerations for program planning

Introduction

[View in ASL](#)

Ontario elementary and secondary schools strive to support high-quality learning and student well-being. Schools give individual students the opportunity to learn in ways that are best suited to their individual strengths and needs. At the secondary level, students' ability to thrive academically and personally is also supported by their ability to choose courses and programs that best suit their skills, interests, and preferred postsecondary destinations.

Educators plan teaching and learning in every subject and discipline so that the various needs of all students are addressed and so that students can see themselves reflected in classroom resources and activities. This section highlights the key strategies and policies that educators and school leaders consider as they plan effective and inclusive programs for all students.

Roles and Responsibilities

Students

Students' responsibilities with respect to their own learning develop gradually and increase over time as they mature and progress through elementary and secondary school. With experience and support, including culturally responsive and relevant instruction, students can:

- look for and act on new ideas and opportunities for learning, communicating with teachers and peers, expressing their voice, and developing their agency to become lifelong learners;
- reflect on their learning, then receive, provide, and respond to meaningful feedback, and set their own goals;
- take responsibility and organize themselves so that they can engage in learning both collaboratively with peers and independently.

Students may experience a variety of circumstances that can make school challenging. The attention, patience, and encouragement shown by teachers can be extremely important in supporting all students in their learning and progress.

Students with special education needs must have an Individual Education Plan (IEP). An IEP is a written plan describing the special education program and/or services required by a particular student, based on a thorough assessment of the strengths and needs that affect the student's ability to learn and to

demonstrate learning. It is expected (and required, if the student is sixteen years of age or older) that secondary students will be given the opportunity to provide input in the development of the IEP. However, any student for whom an IEP is being developed should be consulted to the degree possible.

Parents

Parents¹ play a vital role in education. Studies show that students perform better in school if their parents are involved in their education. Schools offer a variety of parent engagement opportunities, such as providing parents with information about how to support their children's learning at home and at school.

Throughout the Ontario curriculum, students are provided with opportunities for hands-on, experiential learning. To ensure students' health and safety as they learn, parents should inform teachers of any medical conditions that their children have, including allergies, and they should also encourage and remind their children to go to school prepared to participate safely in all activities.

Parents of students with special education needs have an important role to play in the promotion of their child's success at school. This includes participating in the Identification Placement Review Committee (IPRC) process, (if their child is being considered for an IPRC), parent-teacher conferences, and other relevant school activities, as well as participating in the development of their child's Individual Education Plan (IEP).

Further information on the role of parents in their children's education can be found in the following ministry documents [A Parent Engagement Policy for Ontario Schools, 2010](#), [The Ontario Curriculum Review and Revision Guide](#), and other [parent resources](#).

Teachers

Teachers are critical to the success of students. Teachers are responsible for planning and implementing learning and assessment activities that use appropriate, high-impact instructional strategies, including culturally responsive and relevant pedagogy, to support students in achieving the curriculum expectations. Teachers are responsible for understanding and addressing individual students' needs, strengths, and interests, and for ensuring equitable learning opportunities for every student, including through the use of varied teaching, assessment, and evaluation strategies and approaches.

Teachers provide numerous hands-on opportunities for students to develop and refine their knowledge and skills. The learning activities they design should be relevant and authentic for all students, so that

¹ The word *parent(s)* is used on this website to refer to parent(s) and guardian(s). It may also be taken to include caregivers or close family members who are responsible for raising the child.

students can apply their learning to life beyond the school. Equitable opportunities to relate lived experiences, knowledge, and skills to wider contexts motivate students to learn in meaningful ways and to become lifelong learners.

To increase their comfort level and skill, and to ensure effective delivery of the curriculum, teachers should reflect on their own attitudes, biases, and values with respect to the topics they are teaching and seek out current resources, mentors, and professional development and training opportunities, as appropriate.

Teachers are responsible for ensuring the health and safety of every student and for encouraging and motivating students to be mindful of their own health and safety and that of others.

As part of effective teaching practice, teachers communicate with parents throughout the school year, using various methods that may be either formal or informal, to meet the diverse needs of families. Teachers discuss with parents what their children are learning at school, and work to better understand students' experiences outside of school. Ongoing reciprocal communication enables parents to work in partnership with the school, leading to stronger connections between the home and school that foster and support equitable student learning and achievement.

Teachers also have an important role in the success of students with special education needs, including reviewing and updating the IEP as well as implementing effective programs and/or services for students with special education needs.

Principals

The principal works in partnership with teachers and parents to ensure that all students have equitable access to the best possible educational experience. The principal is also a community builder who creates an environment that is welcoming to all, and who ensures that every member of the school community is kept well informed. To support student learning, principals ensure that the Ontario curriculum is being effectively implemented in every classroom using a variety of instructional approaches, and that appropriate resources are made available for teachers and students. To enhance teaching and student learning in all subjects, principals promote professional learning and work with staff to facilitate professional development activities that deepen teachers' knowledge of the curriculum.

Principals ensure the development, implementation, and review of a student's Individual Education Plan (IEP); ensure that parents are consulted in the development of their child's IEP and that they are provided with a copy of the IEP; and ensure that the program is delivered as set out in the IEP.

Community Partners

Community partners are an important resource for schools and students, modelling how the knowledge and skills acquired through the curriculum relate to life beyond the school. As mentors supporting students' experiential learning, community partners can enrich their educational experience, as well as the life of the community.

Community groups can support educators in recruiting practising experts to provide in-class, virtual, or off-site learning experiences for students on topics, concepts, and skills in the curriculum. Schools and school boards can coordinate efforts with community partners and can involve community volunteers in supporting instruction and school and community engagement events held at the school (such as parent engagement nights, art and talent shows, athletic and other extra-curricular activities, science fairs, and technological skills competitions). School boards can collaborate with leaders of community programs, including programs offered through community centres, conservation authorities, libraries, cultural centres, youth-led organizations, and museums.

Community partners are especially relevant for students with special education needs as they frequently bring knowledge and experience from outside the school day and experience. They can support parents and share information and observations about the student's behaviour and learning in a variety of settings and help the school team to develop a common understanding of the student's strengths and needs as they affect the student's ability to learn and demonstrate learning. In addition, members of a cultural community may hold important knowledge and skills related to culturally responsive and relevant teaching and learning. Educators are encouraged to consult with the members of their diverse community to ensure that material and content are culturally responsive and relevant. As an example, schools can contact their board's Indigenous education lead to learn more about representation from various local First Nations, Métis, and Inuit community groups and service delivery organizations in their learning programs.

Student Well-Being and Mental Health

[View in ASL](#)

Promoting the healthy development of all students, as well as enabling all students to reach their full potential, is a priority for educators across Ontario. Students' health and well-being contribute to their ability to learn in all disciplines, and that learning in turn contributes to their overall well-being. A well-rounded educational experience prioritizes well-being and academic success for all students by promoting physical and mental health, social-emotional learning, and inclusion. Parents, community partners, and educators all play critical roles in creating this educational experience.

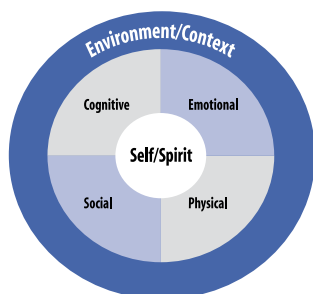
Educators support the well-being of children and youth by creating, fostering, and sustaining a learning environment that is healthy, caring, safe, inclusive, and accepting. A learning environment of this kind supports not only students' cognitive, emotional, social, and physical development but also their sense

of self and/or spirit, their mental health, their resilience, and their overall state of well-being. All this will help them achieve their full potential in school and in life.

A variety of factors, known as “determinants of health”, have been shown to affect a person’s overall state of well-being. Some of these are income, education and literacy, gender and culture, physical and social environment, personal health practices and coping skills, and availability of health services. Together, these factors influence not only whether individuals are physically healthy but also the extent to which they will have the physical, social, and personal resources needed to cope and to identify and achieve personal aspirations. These factors also have an impact on student learning, and it is important to be aware of them as factors contributing to a student’s performance and well-being.

An educator’s awareness of and responsiveness to students’ cognitive, emotional, social, and physical development, and to their sense of self and/or spirit, is critical to their success in school. A number of research-based frameworks, including those described in *Early Learning for Every Child Today: A Framework for Ontario Early Childhood Settings, 2007*, [*On My Way: A Guide to Support Middle Years Childhood Development, 2017*](#), and [*Stepping Stones: A Resource on Youth Development, 2012*](#), identify developmental stages that are common to the majority of students from Kindergarten to Grade 12. At the same time, these frameworks recognize that individual differences, as well as differences in life experiences and exposure to opportunities, can affect development, and that developmental events are not specifically age dependent.

The framework described in *Stepping Stones* is based on a model that illustrates the complexity of human development. Its components – the cognitive, emotional, physical, and social domains – are interrelated and interdependent, and all are subject to the influence of a person’s environment or context. At the centre is an “enduring (yet changing) core” – a sense of self, and/or spirit – that connects the different aspects of development and experience (p. 17).



Source: *Stepping Stones: A Resource on Youth Development*, p. 17

Educators who have an awareness of a student’s development are taking all of the components into account. They focus on the following elements of each component:

- **cognitive development** – brain development, processing and reasoning skills, use of strategies for learning

- **emotional development** – emotional regulation, empathy, motivation
- **social development** – self-development (self-concept, self-efficacy, self-esteem); identity formation (gender identity, social group identity, spiritual identity); relationships (peer, family, romantic)
- **physical development** – physical activity, sleep patterns, changes that come with puberty, body image, nutritional requirements

The Role of Mental Health and Well-Being

Mental health and well-being touch all components of development. Mental health is much more than the absence of mental illness. Well-being depends not only on the absence of problems and risks but also on the presence of factors that contribute to healthy growth and development. By nurturing and supporting students' strengths and assets, educators help promote positive mental health and well-being in the classroom. At the same time, they can identify students who need additional support and connect them with the appropriate supports and services.

What happens at school can have a significant influence on a student's overall well-being. With a broader awareness of mental health, educators can plan instructional strategies that contribute to a supportive classroom climate for learning in all subject areas, build awareness of mental health, and reduce stigma associated with mental illness. Taking students' well-being, including their mental health, into account when planning instructional approaches helps establish a strong foundation for learning and sets students up for success.

Instructional Approaches

[View in ASL](#)

Effective instruction is key to student success. To provide effective instruction, teachers need to consider what they want students to learn, how they will know whether students have learned it, how they will design instruction to promote the learning, and how they will respond to students who are not making progress.

When planning what students will learn, teachers identify the main concepts and skills described in the curriculum expectations, consider the contexts in which students will apply the learning, and determine students' learning goals.

Instructional approaches should be informed by evidence from current research about instructional practices that are effective in the classroom. For example, research has provided compelling evidence about the benefits of explicitly teaching strategies that can help students develop a deeper understanding of concepts. Strategies such as "compare and contrast" (e.g., through Venn diagrams and comparison matrices) and the use of analogy enable students to examine concepts in ways that help

them see what the concepts *are* and what they *are not*. Although such strategies are simple to use, teaching them explicitly is important in order to ensure that all students use them effectively.

A well-planned instructional program should always be at the student's level, but it should also push the student towards their optimal level of challenge for learning, while providing support and anticipating and directly teaching skills that are required for success.

A Differentiated Approach to Teaching and Learning

A differentiated approach to teaching and learning is an important part of a framework for effective classroom practice. It involves adapting instruction and assessment to suit individual students' interests, learning preferences, and readiness in order to promote learning.

An understanding of students' strengths and needs, as well as of their backgrounds, life experiences, and possible emotional vulnerabilities, can help teachers identify and address the diverse strengths and needs of their students. Teachers continually build their awareness of students' learning strengths and needs by observing and assessing their readiness to learn, their interests, and their learning styles and preferences. As teachers develop and deepen their understanding of individual students, they can respond more effectively to each student's needs by differentiating instructional approaches – for example, by adjusting the method or pace of instruction, using different types of resources, allowing a wider choice of topics, or even adjusting the learning environment, if appropriate, to suit the way the student learns and how the student is best able to demonstrate learning. Differentiation is planned as part of the overall learning design, but it also includes making adaptations during the teaching and learning process based on “assessment for learning”. Common classroom strategies that support differentiated instruction include cooperative learning, project-based approaches, problem-based approaches, and explicit instruction. Unless students have an Individual Education Plan with modified expectations, *what* they learn continues to be guided by the curriculum expectations and is the same for all students.

Lesson Design

Effective lesson design involves several important elements. Teachers engage students in a lesson by activating their prior learning and experiences, clarifying the purpose for learning, and making connections to contexts that will help them see the relevance and usefulness of what they are learning. Teachers select instructional strategies to effectively introduce concepts, and consider how they will scaffold instruction in ways that will best meet the needs of their students. At the same time, they consider when and how to check students' understanding and to assess their progress towards achieving their learning goals. Teachers provide multiple opportunities for students to apply their knowledge and skills and to consolidate and reflect on their learning. A three-part lesson design (e.g., “Minds On, Action, and Consolidation”) is often used to structure these elements. Effective lesson design also incorporates culturally responsive and relevant pedagogy (CRRP), which recognizes that all students learn in ways that are connected to background, language, family structure, and social or cultural identity. CRRP is discussed more fully in the section [Equity and Inclusive Education](#).

Planning for Students with Special Education Needs

[View in ASL](#)

Classroom teachers are the key educators of students with special education needs. They have a responsibility to help *all* students learn, and they work collaboratively with special education teachers and educational assistants, where appropriate, to achieve this goal. Classroom teachers commit to assisting every student to prepare for living with the highest degree of independence possible.

[*Learning for All: A Guide to Effective Assessment and Instruction for All Students, Kindergarten to Grade 12, 2013*](#) describes a set of beliefs, based in research, that should guide program planning for students with special education needs. Teachers planning programs or courses in all disciplines need to pay particular attention to these beliefs, which are as follows:

- All students can succeed.
- Each student has their own unique patterns of learning.
- Successful instructional practices are founded on evidence-based research, tempered by experience.
- Universal design² and differentiated instruction³ are effective and interconnected means of meeting the learning or productivity needs of any group of students.
- Classroom teachers are the key educators for a student's literacy and numeracy development.
- Classroom teachers need the support of the larger community to create a learning environment that supports students with special education needs.
- Fairness is not sameness.

In any given classroom, students may demonstrate a wide range of strengths and needs. Teachers plan programs that are attuned to this diversity and use an integrated process of assessment and instruction that responds to the unique strengths and needs of each student. An approach that combines principles of universal design and differentiated instruction enables educators to provide personalized, precise teaching and learning experiences for all students.

In planning programs or courses for students with special education needs, teachers should begin by examining both the curriculum expectations in the grade or course appropriate for the individual

² The goal of Universal Design for Learning (UDL) is to create a learning environment that is open and accessible to all students, regardless of age, skills, or situation. Instruction based on principles of universal design is flexible and supportive, can be adjusted to meet different student needs, and enables all students to access the curriculum as fully as possible.

³ Differentiated instruction is effective instruction that shapes each student's learning experience in response to the student's particular learning preferences, interests, and readiness to learn. See the section [Instructional Approaches](#) for more information.

student and the student's particular strengths and learning needs to determine which of the following options is appropriate for the student:

- no accommodations⁴ or modified expectations; or
- accommodations only; or
- modified expectations, with the possibility of accommodations; or
- alternative expectations, which are not derived from the curriculum expectations for the grade or course and which constitute alternative programs and/or courses.

If the student requires either accommodations or modified expectations, or both, the relevant information, as described in the following paragraphs, must be recorded in their Individual Education Plan (IEP). More detailed information about planning programs for students with special education needs, including students who require alternative programs⁵ and/or courses, can be found in [*Special Education in Ontario, Kindergarten to Grade 12: Policy and Resource Guide, 2017 \(Draft\)*](#) (referred to hereafter as *Special Education in Ontario, 2017*). For a detailed discussion of the ministry's requirements for IEPs, see Part E of *Special Education in Ontario*.

Students Requiring Accommodations Only

Some students with special education needs are able, with certain "accommodations", to participate in the regular grade or course curriculum and to demonstrate learning independently. Accommodations allow the student with special education needs to access the curriculum without changes to the regular expectations. Any accommodations that are required to facilitate the student's learning must be identified in the student's IEP ([*Special Education in Ontario, 2017, p. E38*](#)). A student's IEP is likely to reflect the same required accommodations for many, or all, subjects or courses.

Providing accommodations to students with special education needs should be the first option considered in program planning. Instruction based on principles of universal design and differentiated instruction focuses on providing accommodations to meet the diverse needs of learners.

There are three types of accommodations:

- *Instructional accommodations* are changes in teaching strategies, including styles of presentation, methods of organization, or use of technology and multimedia. Some examples include the use of graphic organizers, photocopied notes, adaptive equipment, or assistive software.
- *Environmental accommodations* are changes that the student may require in the classroom and/or school environment, such as preferential seating or special lighting.

⁴ "Accommodations" refers to individualized teaching and assessment strategies, human supports, and/or individualized equipment (see *Growing Success: Assessment, Evaluation, and Reporting in Ontario Schools, First Edition, Covering Grades 1 to 12, 2010, p. 72*).

⁵ Alternative programs are identified on the IEP by the term "alternative (ALT)".

- *Assessment accommodations* are changes in assessment procedures that enable the student to demonstrate their learning, such as allowing additional time to complete tests or assignments or permitting oral responses to test questions.

(For more examples, see page E39 of *Special Education in Ontario, 2017*.)

If a student requires “accommodations only”, assessment and evaluation of their achievement will be based on the regular grade or course curriculum expectations and the achievement levels outlined for the particular curriculum. The IEP box on the student’s Provincial Report Card will not be checked, and no information on the provision of accommodations will be included.

Students Requiring Modified Expectations

Modified expectations for many students with special education needs will be based on the regular grade or course expectations, with changes in the number and/or complexity of the expectations. Modified expectations must represent specific, realistic, observable, and measurable goals, and must describe specific knowledge and/or skills that the student can demonstrate independently, given the appropriate assessment accommodations.

It is important to monitor, and to reflect clearly in the student’s IEP, the extent to which expectations have been modified. At the secondary level, the principal will determine whether achievement of the modified expectations constitutes successful completion of the course, and will decide whether the student is eligible to receive a credit for the course. This decision must be communicated to the parents and the student.

Modified expectations must indicate the knowledge and/or skills that the student is expected to demonstrate and that will be assessed in each reporting period (*Special Education in Ontario, 2017*, p. E27). Modified expectations should be expressed in such a way that the student and parents can understand not only exactly what the student is expected to know or be able to demonstrate independently, but also the basis on which the student’s performance will be evaluated, resulting in a grade or mark that is recorded on the Provincial Report Card. The student’s learning expectations must be reviewed in relation to the student’s progress at least once every reporting period, and must be updated as necessary (*Special Education in Ontario, 2017*, p. E28).

If a student requires modified expectations, assessment and evaluation of their achievement will be based on the learning expectations identified in the IEP and on the achievement levels outlined under [Levels of Achievement](#) in the “Assessment and Evaluation” section.

Elementary: The IEP box on the Elementary Progress Report Card and the Elementary Provincial Report Card must be checked for any subject in which the student requires modified expectations, and, on the Elementary Provincial Report Card, the appropriate statement from *Growing Success: Assessment, Evaluation, and Reporting in Ontario Schools, First Edition, Covering Grades 1 to 12, 2010*, page 61, must be inserted.

Secondary: If some of the student’s learning expectations for a course are modified but the student is working towards a credit for the course, it is sufficient simply to check the IEP box on the Provincial Report Card, Grades 9–12. If, however, the student’s learning expectations are modified to such an extent that the principal deems that a credit will not be granted for the course, the IEP box must be checked and the appropriate statement from *Growing Success: Assessment, Evaluation, and Reporting in Ontario Schools, First Edition, Covering Grades 1 to 12, 2010*, pages 62–63, must be inserted.

In both the elementary and secondary panels, the teacher’s comments should include relevant information on the student’s demonstrated learning of the modified expectations, as well as next steps for the student’s learning in the subject or course.

Planning for English Language Learners

[View in ASL](#)

English Language Learners in Ontario Schools

Ontario schools have some of the most multilingual student populations in the world. The first language of approximately 28 per cent of the students in Ontario’s English-language schools is a language other than English. In addition, some students use varieties of English – sometimes referred to as dialects – that differ significantly from the English required for success in Ontario schools. Many English language learners were born in Canada and have been raised in families and communities in which languages other than English, or varieties of English that differ from the language used in the classroom, are spoken. Other English language learners arrive in Ontario as newcomers from other countries; they may have experience of highly sophisticated educational systems, or they may have come from regions where access to formal schooling was limited.

When they start school in Ontario, many of these students are entering a new linguistic and cultural environment. All teachers share in the responsibility for these students’ English-language development.

As students who are learning English as a second or additional language in English-language schools, English language learners bring a rich diversity of background knowledge and experience to the classroom. These students’ linguistic and cultural backgrounds not only support their learning in their new environment but also become a cultural asset in the classroom community. Effective teachers find positive ways to incorporate this diversity into their instructional programs and into the classroom environment.

Most English language learners in Ontario schools have age-appropriate proficiency in their first language, as well as age-appropriate literacy skills. Although they need frequent opportunities to use English at school, they also derive important educational and social benefits from continuing to develop their first language while they are learning English. Teachers should encourage parents to continue to use their own language at home, both to preserve the language as part of their children’s heritage and

identity and to provide a foundation for their language and literacy development in English. It is also important for teachers to find opportunities to bring students' languages into the classroom, using parents and community members as a resource.

English as a Second Language and English Literacy Development Programs

During their first few years in Ontario schools, English language learners may receive support through one of two distinct programs designed to meet their language-learning needs:

English as a Second Language (ESL) programs are for students born in Canada or newcomers whose first language is a language other than English, or is a variety of English significantly different from that used for instruction in Ontario schools. Students in these programs have had educational opportunities to develop age-appropriate first-language literacy skills.

English Literacy Development (ELD) programs are primarily for newcomers whose first language is a language other than English, or is a variety of English significantly different from that used for instruction in Ontario schools, and who arrive with significant gaps in their education. These students generally come from countries where access to education is limited or where there are limited opportunities to develop language and literacy skills in any language. Schooling in their countries of origin may have been inconsistent, disrupted, or even completely unavailable throughout the years that these children would otherwise have been in school.

Supportive Learning Environments

In planning programs for students with linguistic backgrounds other than English, teachers need to recognize the importance of the orientation process, understanding that every learner needs to adjust to the new social environment and language in a unique way and at an individual pace. For example, students who are in an early stage of English-language acquisition may go through a "silent period" during which they closely observe the interactions and physical surroundings of their new learning environment. They may use body language rather than speech or they may use their first language until they have gained enough proficiency in English to feel confident of their interpretations and responses. Students thrive in a safe, supportive, and welcoming environment that nurtures their self-confidence while they are receiving focused literacy instruction. When they are ready to participate, in paired, small-group, or whole-class activities, some students will begin by using a single word or phrase to communicate a thought, while others will speak quite fluently.

In a supportive learning environment, most students will develop oral language proficiency quite quickly. Teachers can sometimes be misled by the high degree of oral proficiency demonstrated by many English language learners in their use of everyday English and may mistakenly conclude that these students are equally proficient in their use of academic English. Most English language learners who have developed oral proficiency in everyday English will still require instructional scaffolding to meet curriculum expectations. Research has shown that it takes five to seven years for most English language learners to catch up to their English-speaking peers in their ability to use English for academic purposes.

Program Adaptations

Responsibility for students' English-language development is shared by all teachers, including the ESL/ELD teacher (where available), and other school staff. Volunteers and peers may also be helpful in supporting English language learners in the classroom. By adapting the instructional program, teachers facilitate these students' learning. Appropriate adaptations include modifications and accommodations, as follows:

- modification of some or all of the grade or course expectations so that they are challenging but attainable for the learners at their current level of English proficiency, with the necessary support from the teacher;
- use of a variety of instructional strategies;⁶
- use of a variety of learning resources;⁷
- use of assessment accommodations that support students in demonstrating the full range of their learning.⁸

Teachers need to adapt the program for English language learners as they acquire English proficiency. For English language learners at the early stages of English language acquisition, teachers are required to modify curriculum expectations as needed. Most English language learners require accommodations for an extended period, long after they have achieved proficiency in everyday English.

Assessment and Evaluation

When curriculum expectations are modified in order to meet the language-learning needs of English language learners, assessment and evaluation will be based on the documented modified expectations. Teachers will check the ESL/ELD box on the Provincial Report Card only when modifications have been made to curriculum expectations to address the language needs of English language learners (the box

⁶ Examples include: small-group instruction; extensive use of visual cues, images, diagrams; visual representations of key ideas; graphic organizers; scaffolding; previewing of text; modelling; use of music, movement, and gestures; open-ended activities; pre-teaching of key vocabulary; peer tutoring; strategic use of students' first languages.

⁷ Examples include: visual material; simplified text; bilingual dictionaries; subject-specific glossaries; resources available in languages that students speak at home; concrete materials; learning materials and activities – displays, music, dances, games, and so on – that reflect cultural diversity.

⁸ Examples include: provision of additional time; provision of options for students to choose how they will demonstrate their learning, such as portfolios, oral interviews, presentations, oral or visual representations, demonstrations and models, dramatic activities, and songs and chants; use of tasks requiring completion of graphic organizers or cloze sentences instead of essay questions or other assessment tasks that depend heavily on proficiency in English.

should *not* be checked to indicate simply that they are participating in ESL/ELD programs or if they are only receiving accommodations). There is no requirement for a statement to be added to the “Comments” section of the report cards when the ESL/ELD box is checked.

Although the degree of program adaptation required will decrease over time, students who are no longer receiving ESL or ELD support may still need some program adaptations to be successful.

Related Policy and Resource Documents

For further information on supporting English language learners, refer to the following documents:

- [*Growing Success: Assessment, Evaluation, and Reporting in Ontario Schools, First Edition, Covering Grades 1 to 12, 2010*](#)
- [*The Ontario Curriculum, Grades 9–12: English as a Second Language and English Literacy Development, 2007*](#)
- [*English Language Learners – ESL and ELD Programs and Services: Policies and Procedures for Ontario Elementary and Secondary Schools, Kindergarten to Grade 12, 2007*](#)
- [*Supporting English Language Learners with Limited Prior Schooling: A Practical Guide for Ontario Educators, Grades 3 to 12, 2008*](#)
- [*Supporting English Language Learners: A Practical Guide for Ontario Educators, Grades 1 to 8, 2008*](#)
- [*Many Roots, Many Voices: Supporting English Language Learners in Every Classroom, 2005.*](#)

Healthy Relationships

[View in ASL](#)

Every student is entitled to learn in a safe, caring environment, free from discrimination, violence, and harassment. Research has shown that students learn and achieve better in such environments. A safe and supportive social environment in a school is founded on healthy relationships – the relationships between students, between students and adults, and between adults. Healthy relationships are based on respect, caring, empathy, trust, and dignity, and thrive in an environment in which diversity is honoured and accepted. Healthy relationships do not tolerate abusive, controlling, violent, bullying/harassing, or other inappropriate behaviours. To experience themselves as valued and connected members of an inclusive social environment, students need to be involved in healthy relationships with their peers, educators, and other members of the school community.

Several provincial policies, programs, and initiatives, including [Foundations for a Healthy School](#), the [Equity and Inclusive Education Strategy](#), and [Safe Schools](#), are designed to foster caring and safe learning environments in the context of healthy and inclusive schools. These policies and initiatives promote positive learning and teaching environments that support the development of healthy relationships, encourage academic achievement, and help all students reach their full potential.

In its 2008 report, [*Shaping a Culture of Respect in Our Schools: Promoting Safe and Healthy Relationships*](#), the Safe Schools Action Team confirmed “that the most effective way to enable all students to learn about healthy and respectful relationships is through the school curriculum” (p. 11). Educators can promote this learning in a variety of ways. For example, by giving students opportunities to apply critical thinking and problem-solving strategies and to address issues through group discussions, role play, case study analysis, and other means, they can help them develop and practise the skills they need for building healthy relationships. Co-curricular activities such as clubs and intramural and interschool sports provide additional opportunities for the kind of interaction that helps students build healthy relationships. Educators can also have a positive influence on students by modelling the behaviours, values, and skills that are needed to develop and sustain healthy relationships, and by taking advantage of “teachable moments” to address immediate relationship issues that may arise among students.

Human Rights, Equity, and Inclusive Education

[View in ASL](#)

A positive, inclusive, equitable, and non-discriminatory elementary and secondary school experience is vitally important to a student’s personal, social, and academic development, to their future economic security, and to a realization of their full potential. Human rights principles recognize the importance of creating a climate of understanding and mutual respect for the dignity and worth of each person, so that each person can contribute fully to the development and well-being of their community. Indeed, human rights law guarantees a person’s right to equal treatment in education. It requires educators and school leaders to prevent and respond appropriately to discrimination and harassment, to create an inclusive environment, to remove barriers that limit the ability of students, and to provide accommodations, where necessary.

Ontario’s education system, at all levels, must respect diversity, promote inclusive education, and work towards identifying and eliminating barriers to equal treatment in education that limit the ability of students to learn, grow, and contribute to society. Discriminatory biases, harassment, non-inclusive environments, lack of accommodation, systemic barriers, power dynamics, societal poverty, and racism make it difficult for students to acquire the skills they need to be successful, competitive, and productive members of society. Ontario schools aim to improve the academic outcomes and experiences of students who have traditionally not benefited from the promise of public education.

In an environment based on the principles of inclusive education, all students, parents, caregivers, and other members of the school community – regardless of ancestry, culture, ethnicity, sex, disability, race, colour, religion, age, marital or family status, creed, gender identity/expression, gender, sexual orientation, socio-economic status, or other factors – are welcomed, included, treated fairly, and respected. Diversity is valued when all members of the school community feel safe, welcomed, and accepted. Every student is supported and inspired to succeed in a culture of high expectations for learning.

Research has shown that students who do not see themselves reflected in what they are learning, in their classrooms, and in their schools become disengaged and do not experience as great a sense of well-being or as high a level of academic achievement as those who do.

Culturally Responsive and Relevant Pedagogy (CRRP)

In an inclusive education system, students must see themselves reflected in the curriculum, their physical surroundings, and the broader environment, so that they can feel engaged in and empowered by their learning experiences. Students need to experience teaching and learning that reflect their needs and who they are. To ensure that this happens, educators in Ontario schools embrace *culturally responsive and relevant pedagogy* (CRRP), which recognizes that all students learn in ways that are connected to background, language, family structure, and social or cultural identity.

CRRP provides a framework for building positive environments, improving student responsibility and success, encouraging parent-school relationships, and building strong community connections. It also emphasizes that it is important for educators and school leaders to examine their own biases and to analyse how their own identities and experiences affect how they view, understand, and interact with all students. This can help to prevent discrimination, harassment, and the creation of poisoned environments. Educators are responsible for meaningful teaching and learning that recognizes and responds to *who is in the classroom and the school*.

By knowing “who our students are”, educators and leaders can tailor policies, programs, and practices to better meet the needs of their diverse student populations, to provide accommodation of the needs specified by human rights law, and to ensure that every student has the opportunity to succeed. CRRP involves recognizing that “culture” encompasses various aspects of social and personal identity. It also means acknowledging students’ multiple social and personal identities and the social issues that arise where identities intersect. The CRRP approach is designed to spark conversation and support educators and school leaders as they seek to implement effective equity strategies and policies. Educators are encouraged to engage in meaningful inquiry, in collaboration with colleagues, to address equity issues and the particular needs of the students they serve.

Implementing Principles of Inclusive Education

The implementation of inclusive education principles in education influences all aspects of school life. It promotes a school climate that encourages all students to work to high levels of achievement, affirms the worth of all students, and helps students strengthen their sense of identity and develop a positive self-image. It encourages staff and students alike to value and show respect for diversity in the school and the broader society. Inclusive education promotes equity, healthy relationships, and active, responsible citizenship. The absence of inclusive approaches to education can create discriminatory environments, in which certain individuals or groups cannot expect to receive fair treatment or an equitable experience based on aspects of their identity.

Teachers can give students a variety of opportunities to learn about diversity and diverse perspectives. By drawing attention to the contributions and perspectives of historically marginalized groups, and by

creating opportunities for their experiences to be affirmed and valued, teachers can enable students from a wide range of backgrounds to see themselves reflected in the curriculum. It is essential that learning activities and materials used to support the curriculum reflect the diversity of Ontario society. In addition, teachers should differentiate instruction and assessment strategies to take into account the background and experiences, as well as the interests, aptitudes, and learning needs, of all students.

Interactions between the school and the community should reflect the diversity of both the local community and the broader society. A variety of strategies can be used to communicate with and engage parents and members of diverse communities, and to encourage their participation in and support for school activities, programs, and events. Family and community members should be invited to take part in teacher interviews, the school council, and the parent involvement committee, and to attend and support activities such as plays, concerts, co-curricular activities and events, and various special events at the school. Schools need to be prepared and ready to welcome families and community members. Schools may consider offering assistance with child care or making alternative scheduling arrangements in order to help caregivers participate. Special outreach strategies and encouragement may be needed to draw in the parents of English language learners and First Nations, Métis, or Inuit students, and to make them feel more welcomed in their interactions with the school.

The Role of the School Library

[View in ASL](#)

The school library program can help build and transform students' knowledge in order to support lifelong learning in our information- and knowledge-based society. The school library program supports student success across the curriculum by encouraging students to read widely, teaching them to examine and read many forms of text for understanding and enjoyment, and helping them improve their research skills and effectively use information gathered through research.

The school library program enables students to:

- develop a love of reading for learning and for pleasure;
- develop literacy skills using fiction and non-fiction materials;
- develop the skills to become independent, thoughtful, and critical researchers;
- obtain access to programs, resources, and integrated technologies that support all curriculum areas;
- understand and value the role of public library systems as a resource for lifelong learning.

The school library program plays a key role in the development of information literacy and research skills. Teacher-librarians, where available, collaborate with classroom or content-area teachers to design, teach, and provide students with authentic information and research tasks that foster learning, including the ability to:

- access, select, gather, process, critically evaluate, create, and communicate information;

- use the information obtained to explore and investigate issues, solve problems, make decisions, build knowledge, create personal meaning, and enrich their lives;
- communicate their findings to different audiences, using a variety of formats and technologies;
- use information and research with understanding, responsibility, and imagination.

In addition, teacher-librarians can work with content-area teachers to help students:

- develop digital literacy in using non-print forms, such as the Internet, social media, and blogs, and knowing the best ways to access relevant and reliable information;
- design inquiry questions for research projects;
- create and produce single-medium or multimedia presentations.

Teachers need to discuss with students the concept of ownership of work and the importance of copyright in all forms of media.

The Role of Information and Communications Technology

[View in ASL](#)

The variety and range of information and communications technology (ICT) tools available to educators today enables them to significantly extend and enrich their instructional approaches and to create opportunities for students to learn in ways that best suit their interests and strengths. Technology has also enhanced the ability to connect with communities outside the school, making it possible to engage a diversity of community partners in student learning.

Rich opportunities can be tapped to support students in developing [digital literacy](#), an essential transferable skill.

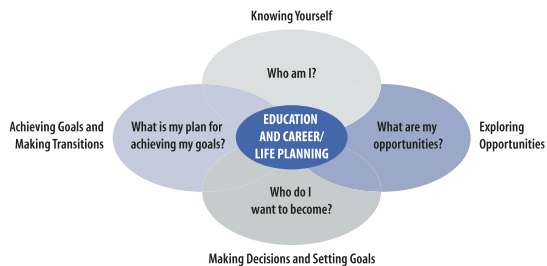
Education and Career/Life Planning

[View in ASL](#)

The goals of the Kindergarten to Grade 12 education and career/life planning program are to:

- ensure that all students develop the knowledge and skills they need to make informed education and career/life choices;
- provide classroom and school-wide opportunities for this learning; and
- engage parents and the broader community in the development, implementation, and evaluation of the program, to support students in their learning.

The framework of the program is a four-step inquiry process based on four questions linked to four areas of learning: (1) Knowing Yourself – Who am I?; (2) Exploring Opportunities – What are my opportunities?; (3) Making Decisions and Setting Goals – Who do I want to become?; and (4) Achieving Goals and Making Transitions – What is my plan for achieving my goals?



The curriculum expectations in most subjects and disciplines of the Ontario curriculum provide opportunities to relate classroom learning to the education and career/life planning program as outlined in [*Creating Pathways to Success: An Education and Career/Life Planning Program for Ontario Schools – Policy and Program Requirements, Kindergarten to Grade 12, 2013*](#). All classroom teachers support students in education and career/life planning by providing them with learning opportunities, filtered through the lens of the four inquiry questions, that allow them to reflect on and apply subject-specific knowledge and skills; explore subject-related education and career/life options; and become competent, self-directed planners who will be prepared for success in school, life, and work. Education and career/life planning will support students in their transition from secondary school to their initial postsecondary destination, whether it be in apprenticeship training, college, community living, university, or the workplace. For more information on postsecondary pathway choices, see the [Education and Training](#) and [Skilled Trades](#) pages on the Ontario government website.

Experiential Learning

[View in ASL](#)

Experiential learning is hands-on learning that occurs in person or virtually and provides developmentally appropriate opportunities for students of all ages to:

- **participate** in rich experiences connected to the world outside the school;
- **reflect** on the experiences to derive meaning; and
- **apply** the learning to their decisions and actions.

Adapted from David A. Kolb, *Experiential Learning: Experience as the Source of Learning and Development*, 2nd ed. (Upper Saddle River, N.J.: Pearson Education, 2015)

Planned learning experiences in the community may include outdoor education, project/program-based learning, job shadowing and job twinning, field trips, field studies, work experience, and cooperative education. These experiences provide opportunities for students to see the relevance of their classroom learning and its connection to the broader world. They also help them develop transferable and interpersonal skills and work habits that prepare them for their future, and enable them to explore careers of interest as they plan their pathway through school to their postsecondary destination, whether in apprenticeship training, college, community living, university, or the workplace.

Experiential learning opportunities associated with various aspects of the curriculum help broaden students' knowledge of themselves and of a range of career opportunities – two areas of learning outlined in [*Creating Pathways to Success: An Education and Career/Life Planning Program for Ontario Schools – Policy and Program Requirements, Kindergarten to Grade 12, 2013*](#). The key to providing successful experiential learning opportunities is to ensure that the experiential learning cycle (participate, reflect, apply) is a planned part of the experience.

In secondary school, pathways programs that incorporate experiential learning are available to students. They include the following courses and programs:

- cooperative education courses, outlined in [*The Ontario Curriculum, Grades 11–12: Cooperative Education, 2018*](#)
- Ontario Youth Apprenticeship Program (OYAP) (see [“Prepare for Apprenticeship”](#) on the Ontario government website)
- [Specialist High Skills Major \(SHSM\)](#) program
- [Dual credit](#) programs

Pathways to a Specialist High Skills Major (SHSM)

[View in ASL](#)

The [Specialist High Skills Major \(SHSM\)](#) is a specialized, ministry-approved program that allows students in Grades 11 and 12 to focus their learning on a specific economic sector while meeting the requirements of the Ontario Secondary School Diploma (OSSD).

The SHSM program assists students in their transition from secondary school to apprenticeship training, college, university, or the workplace.

This program enables students to gain sector-specific skills and knowledge in engaging, career-related learning environments and to prepare in a focused way for graduation and postsecondary education, training, or employment.

Course offerings and program planning should support students who are pursuing specialized programs, including the SHSM program. Bundles of credits provide students with knowledge and skills that are

connected with the specific sector of their SHSM program and that are required for success in their chosen destination.

Health and Safety

[View in ASL](#)

In Ontario, various laws, including the [Education Act](#), the [Occupational Health and Safety Act](#) (OHSA), [Ryan's Law \(Ensuring Asthma Friendly Schools\), 2015](#), and [Sabrina's Law, 2005](#), collectively ensure that school boards provide a safe and productive learning and work environment for both students and employees. Under the Education Act, teachers are required to ensure that all reasonable safety procedures are carried out in courses and activities for which they are responsible. Teachers should model safe practices at all times; communicate safety requirements to students in accordance with school board policies, Ministry of Education policies, and any applicable laws; and encourage students to assume responsibility for their own safety and the safety of others.

Concern for safety should be an integral part of instructional planning and implementation. Teachers are encouraged to review:

- their responsibilities under the [Education Act](#);
- their rights and responsibilities under the [Occupational Health and Safety Act](#);
- their school board's health and safety policy for employees;
- their school board's policies and procedures on student health and safety (e.g., on concussions; on medical conditions such as asthma; with respect to outdoor education excursions);
- relevant provincial subject association guidelines and standards for student health and safety, such as Ophea's [Ontario Physical Activity Safety Standards in Education](#) (formerly the Ontario Physical Education Safety Guidelines);
- any additional mandatory requirements, particularly for higher-risk activities (e.g., field trips that involve water-based activities), including requirements for approvals (e.g., from the Supervisory Officer), permissions (e.g., from parents/guardians), and/or qualifications (e.g., proof of students' successful completion of a swim test).

Wherever possible, potential risks should be identified and procedures developed to prevent or minimize, and respond to, incidents and injuries. School boards provide and maintain safe facilities and equipment, as well as qualified instruction. In safe learning environments, teachers will:

- be aware of up-to-date safety information;
- plan activities with safety as a primary consideration;
- inform students and parents of risks involved in activities;
- observe students to ensure that safe practices are being followed;
- have a plan in case of emergency;
- show foresight;

- act quickly.

Students should be made aware that health and safety is everyone's responsibility – at home, at school, and in the community. Teachers should ensure that students have the knowledge and skills needed for safe participation in all learning activities. Students must be able to demonstrate knowledge of the equipment being used and the procedures necessary for its safe use. Health and safety resource guides for [Kindergarten to Grade 8](#) and for [Grades 9 to 12](#) provide the scope and sequence of Ontario curriculum expectations to assist teachers in bringing health and safety education into the classroom in every subject area. The guides identify expectations in the Ontario curriculum that can help students develop knowledge and skills related to health and safety (injury prevention and health protection), safe behaviours, and safe practices.

Learning outside the classroom, such as on field trips or during field studies, can provide a meaningful and authentic dimension to students' learning experiences, but they also take the teacher and students out of the predictable classroom environment and into unfamiliar settings. Teachers must plan these activities carefully in accordance with their school board's relevant policies and procedures and in collaboration with other school board staff (e.g., the principal, outdoor education lead, Supervisory Officer) to ensure students' health and safety.

The information provided in this section is not exhaustive. Teachers are expected to follow school board health and safety policies and procedures.

Ethics

[View in ASL](#)

The Ontario curriculum provides varied opportunities for students to learn about ethical issues and to explore the role of ethics in both public and personal decision making. Students may make ethical judgements when evaluating evidence and positions on various issues, and when drawing their own conclusions about issues, developments, and events. Teachers may need to help students determine which factors they should consider when making such judgements. It is crucial that teachers provide support and supervision to students throughout the research and inquiry process, ensuring that students engaged in an inquiry are aware of potential ethical concerns and that they address such concerns in acceptable ways. Teachers may supervise students' use of surveys and/or interviews, for example, to confirm that their planned activities will respect the dignity, privacy, and confidentiality of their participants. When students' activities involve Indigenous communities and/or individuals, teachers need to ensure the appropriate use and protection of Indigenous knowledge. Teachers also supervise the choice of the research topics to protect students from exposure to information and/or perspectives for which they may not be emotionally or intellectually prepared (for example, where a student's investigation might involve personal interviews that could lead to the disclosure of abuse or other sensitive topics).

Teachers must thoroughly address the issues of plagiarism and cultural appropriation with students. In a digital world that provides quick access to abundant information, it is easy to copy the words, music, or images of others and present them as one's own. Even at the secondary level, students need to be reminded of the ethical issues related to plagiarism and appropriation. Before starting an inquiry, students should have an understanding of the range of forms of plagiarism and appropriation, from blatant to nuanced, as well as of their consequences. Students often struggle to find a balance between creating works in their own voice or style and acknowledging the work of others. It is not enough to tell them not to plagiarize or appropriate others' work, and to admonish those who do. Teachers need to explicitly teach all students how to use their own voice or style while appropriately acknowledging the work of others, using accepted forms of documentation.

Cross-curricular and integrated learning

Introduction

[View in ASL](#)

A variety of overarching perspectives, themes, and skills are intentionally incorporated by educators, on an ongoing basis, into teaching and learning across all subjects and disciplines of the curriculum – they are part of “cross-curricular learning”. Educators plan programs to include learning in these areas, which are relevant in the context of most curriculum subjects, and are critical to students in navigating their world. They range from environmental education, Indigenous education, and financial literacy to social-emotional learning, critical literacy, mathematical literacy, and STEM education. These various themes, perspectives, and skills are explored in this section.

Another approach to teaching and learning “across subjects” is called “integrated learning”. This approach differs from cross-curricular learning because it involves combining curriculum expectations from more than one subject in a single lesson, and evaluating student achievement of the expectations within the respective subjects from which they are drawn.

Scope and Sequence Resource Guides

“Scope and sequence” resource guides are compilations of existing curriculum expectations, from all subjects and disciplines, that relate to specific ministry priorities and initiatives. For example, scope and sequence resource guides have been developed for **environmental education** ([elementary](#) and [secondary](#)); **financial literacy** ([elementary](#) and [secondary](#)); **First Nations, Métis, and Inuit connections** ([elementary](#) and [secondary](#)); and **health and safety** ([elementary](#) and [secondary](#)).

These documents identify expectations that involve learning about the particular topic, as well as teacher supports that touch on the topic or that describe opportunities for addressing it. The teacher supports include the examples, sample questions, teacher prompts, student responses, and/or instructional tips that accompany the expectations and describe optional ways in which teachers can elicit the learning described in the expectation. Teachers can glean ideas from the teacher supports, based on their professional judgement and taking into account the interests of the students and the local communities represented in their classrooms, for incorporating learning about these topics across subjects. The scope and sequence resource guides can also support divisional/school planning on particular topics or issues across classrooms and grades.

Integrated Learning

[View in ASL](#)

Integrated learning engages students in a rich learning experience that helps them make connections across subjects and brings the learning to life. Integrated learning provides students with opportunities to work towards meeting expectations from two or more subjects within a single unit, lesson, or activity. It can be a solution to the problems of fragmented learning and isolated skill instruction, because it provides opportunities for students to learn and apply skills in meaningful contexts across subject boundaries. In such contexts, students have opportunities to develop their ability to think and reason and to transfer knowledge and skills from one subject area to another. Although the learning is integrated, *the specific knowledge and skills from the curriculum for each subject are taught.*

Elementary Curriculum

By linking expectations from different subjects within a single unit, lesson, or activity, elementary teachers can provide students with multiple opportunities to reinforce and demonstrate their knowledge and skills in a variety of contexts. Teachers then evaluate student achievement in terms of the individual expectations, towards assigning a grade for each of the subjects involved.

One example would be a unit linking expectations from the science and technology curriculum and from the social studies curriculum. Connections can be made between these curricula in a number of areas – for example, the use of natural resources, considered from a scientific and an economic perspective; variations in habitat and ecosystems across the regions of Canada, exploring both the biology and the geography of those regions; historical changes in technology; and the impact of science and technology on various peoples and on the environment. In addition, a unit combining science and technology and social studies expectations could teach inquiry/research skills common to the two subjects, while also introducing approaches unique to each.

Secondary Curriculum

Ontario's secondary curriculum is designed to provide opportunities for educators to integrate student learning across disciplines and subjects. Some secondary expectations are written to implicitly connect with and support content learning and skill development outlined in other curricula. For example, the secondary math and science curricula are aligned so that students can apply what they learn in math to what they are learning in the sciences. For instance, in Grade 11 and 12 math courses, students learn the mathematical concepts needed to support learning in chemistry and physics courses in those grades. As another example, expectations in social sciences and humanities are aligned with some of the expectations in the English curriculum.

Financial Literacy

[View in ASL](#)

The education system has a vital role to play in preparing young people to take their place as informed, engaged, and knowledgeable citizens in the global economy. Financial literacy education can provide the preparation Ontario students need to make informed decisions and choices in a complex and fast-changing financial world.

Because making informed decisions about economic and financial matters has become an increasingly complex undertaking in the modern world, students need to build knowledge and skills in a wide variety of areas. In addition to learning about the specifics of saving, spending, borrowing, and investing, students need to develop broader skills in problem solving, research and inquiry, decision making, critical thinking, and critical literacy related to financial issues, so that they can analyse and manage the risks that accompany various financial choices. They also need to develop an understanding of world economic forces and the effects of those forces at the local, national, and global level. In order to make wise choices, they will need to understand how such forces affect their own and their families' economic and financial circumstances. Finally, to become responsible citizens in the global economy, they will need to understand the social, environmental, and ethical implications of their own choices as consumers. For all of these reasons, financial literacy is an essential component of the education of Ontario students in a twenty-first century context – one that can help ensure that Ontarians will continue to prosper in the future.

Resource documents – [The Ontario Curriculum, Grades 4–8: Financial Literacy Scope and Sequence of Expectations, 2016](#) and [The Ontario Curriculum, Grades 9–12: Financial Literacy Scope and Sequence of Expectations, 2016](#) – have been prepared to assist teachers in bringing financial literacy into the classroom. These documents identify the curriculum expectations and related examples and prompts, in disciplines across the Ontario curriculum, through which students can acquire skills and knowledge related to financial literacy.

STEM Education

[View in ASL](#)

K–12 STEM education is the study of science, technology, engineering, and mathematics, including cross-curricular and/or integrative study, and the application of those subjects in real-world contexts. As students engage in STEM education, they develop [transferable skills](#) that they need to meet the demands of today’s global economy and society.

STEM education helps students develop an understanding and appreciation of each of the core subjects of mathematics, science, and technological education. At the same time, it supports a more holistic understanding and application of skills and knowledge related to engineering design and innovation. STEM learning integrates and applies concepts, processes, and ways of thinking associated with these subjects to design solutions to real-world problems.

Engineering design and innovation engages students in *applying* the principles of science, technology, and mathematics to develop economical and sustainable solutions to technical and complex societal problems to meet human needs.

Among the transferable skills developed through STEM education are computational thinking, coding, design thinking, innovating, use of the scientific method, scientific inquiry skills, and engineering design skills. These skills are in high demand in today’s globally connected world, with its unprecedented advancements in technology.

Approaches to STEM education may vary across Ontario schools. STEM subjects may be taught separately, but with an effort to make cross-curricular connections a part of student learning. Problem-solving application projects may be designed to combine two or more STEM subjects. Alternatively, content from all four STEM subjects might be fully integrated to reinforce students' understanding of each subject, by enhancing their understanding of the interrelationships among them, and by providing the opportunity to apply a spectrum of knowledge and skills in novel ways in real-world contexts. As STEM education is implemented, it is important to engage diverse perspectives and ways of thinking, including those inherent in the arts and humanities. Diverse perspectives engage students in a variety of creative and critical thinking processes that are essential for developing innovative and effective solutions that impact communities or ecosystems.

A robust K–12 STEM education enables Ontario educators and students to become innovators and leaders of change in society and the workforce, and creates opportunities in our diverse communities to foster integrative thinking and problem solving.

Indigenous Education

[View in ASL](#)

To move forward on their learning journey, students must have a solid understanding of where we have been as a province and as a country. Consistent with Ontario’s vision for Indigenous education, all students will have knowledge of the rich diversity of First Nations, Métis, and Inuit histories, cultures, perspectives, and contributions, as well as an awareness of the importance of Indigenous ways of knowing in a contemporary context. Ontario is committed to ensuring that First Nations, Métis, and Inuit survivors and communities bring their perspectives to students’ learning about our shared history.

It is essential that learning activities and resources used to support Indigenous education are authentic and accurate and do not perpetuate culturally and historically inaccurate ideas and understandings. It is important for educators and schools to select resources that represent the uniqueness of First Nations, Métis, and Inuit histories, perspectives, and world views authentically and respectfully. It is also important to select resources that reflect local Indigenous communities as well as First Nations, Métis, and Inuit individuals and communities from across Ontario and Canada. Resources that best support Indigenous education feature Indigenous voices and narratives and are developed by, or in collaboration with, First Nations, Métis, and Inuit communities. Schools can contact their board’s Indigenous lead and work with their Indigenous Education Councils for assistance in evaluating and selecting resources.

Cultural Safety

It is important to create a learning environment that is respectful and that makes students feel safe and comfortable not only physically, socially, and emotionally but also in terms of their cultural heritage. A culturally safe learning environment is one in which students feel comfortable about expressing their ideas, opinions, and needs and about responding authentically to topics that may be culturally sensitive. Educators should be aware that some students may experience emotional reactions when learning about issues that have affected their own lives, their family, and/or their community, such as the legacy of the residential school system. Before addressing such topics in the classroom, teachers need to consider how to prepare and debrief students, and they need to ensure that appropriate resources are available to support students both inside and outside the classroom.

Literacy

[View in ASL](#)



Literacy is the ability to use language and images in rich and varied forms to read, write, listen, speak, view, represent, discuss, and think critically about ideas. Literacy enables us to share information and to interact with others. Literacy is an essential tool for personal growth and active participation in a democratic society.

- Ontario Ministry of Education, [*Paying Attention to Literacy: Six Foundations for Improvement in Literacy, K-12, 2013*](#)

The Importance of Literacy

Literacy⁹ continues to evolve as the world changes and its demands shift and become more complex. A focus on literacy goes beyond traditional forms of reading and writing. Today's students live with technological innovations that previous generations never experienced. They are accustomed to receiving information quickly, and often in a non-linear format, and they may engage in social interactions using a variety of technologies.

Literacy skills are embedded in the expectations for all subjects and disciplines of the Ontario curriculum. Each subject provides opportunities for literacy development, often in specialized ways. Literacy needs to be explicitly taught in all subjects. Literacy demands, such as vocabulary acquisition and accessing and managing information, become more complex across subjects and disciplines as students progress through the grades.

The Scope of Literacy

In Ontario schools, all students are equipped with the literacy skills necessary to be critical and creative thinkers, effective meaning-makers and communicators, collaborative co-learners, and innovative problem-solvers. These are the skills that will enable them to achieve personal, career, and societal goals. Students develop literacy skills as they think, express, and reflect.

In every subject, before, during, and after they read, view, listen, speak, or write, students select and use a variety of literacy strategies and subject-specific processes. This helps them comprehend and organize information and ideas, and communicate meaning. Teachers assist students in learning and selecting appropriate literacy strategies based on assessment of their individual needs and learning preferences.

Students learn to think, express, and reflect in discipline-specific ways. Teachers purposefully teach students about the literacy demands of the particular subject area. Students learn the vocabulary and terminology that are unique to a particular subject area and must be able to interpret symbols, charts and diagrams. Cross-curricular and subject-specific literacy skills are essential to students' success in all subjects of the curriculum, and in all areas of their lives.

⁹ This page has been adapted from the French resource: [*Guide de la litt  ratie chez les adolescentes et adolescents, Ressource d'apprentissage professionnel en mati  re de litt  ratie, de la 7      la 12   ann  e*](#), Direction des politiques relatives au curriculum et    l'  valuation, minist  re de l'  ducation de l'Ontario (2012).

Critical Thinking and Critical Literacy

[View in ASL](#)

Critical thinking is the process of thinking about ideas or situations in order to understand them fully, identify their implications, make a judgement, and/or guide decision making. It is an essential transferable skill that enables students to become independent, informed, and responsible members of society, and so is a focus of learning across all subjects and disciplines. Critical thinking includes skills such as questioning, predicting, analysing, synthesizing, examining opinions, identifying values and issues, detecting bias, and distinguishing between alternatives. Students who are taught these skills become critical thinkers who can move beyond superficial conclusions to a deeper understanding of the issues they are examining. They are able to engage in an inquiry process in which they explore complex and multifaceted issues, and questions for which there may be no clear-cut answers.

Students use critical-thinking skills when they assess, analyse, and/or evaluate the impact of something and when they form an opinion and support that opinion with a rationale. In order to think critically, students need to ask themselves effective questions in order to interpret information; detect bias in their sources; determine why a source might express a particular bias; examine the opinions, perspectives, and values of various groups and individuals; look for implied meaning; and use the information gathered to form a personal opinion or stance, or a personal plan of action with regard to making a difference.

Students approach critical thinking in various ways. Some students find it helpful to discuss their thinking, asking questions and exploring ideas. Other students may take time to observe a situation or consider a text carefully before commenting; they may prefer not to ask questions or express their thoughts orally while they are thinking.

Critical literacy is the term used to refer to a particular aspect of critical thinking. Critical literacy involves looking beyond the literal meaning of a text to determine what is present and what is missing, in order to analyse and evaluate the text's complete meaning and the author's intent. Critical literacy is concerned with issues related to fairness, equity, and social justice. Critically literate students adopt a critical stance, asking what view of the world the text advances and whether they find this view acceptable, who benefits from the text, and how the reader is influenced.

Critically literate students understand that meaning is not found in texts in isolation. People make sense of a text, or determine what a text means, in a variety of ways. Students therefore need to take into account: points of view (e.g., those of people from various cultures); context (e.g., the beliefs and practices of the time and place in which a text was created and those in which it is being read or viewed); the background of the person who is interacting with the text (e.g., upbringing, friends, communities, education, experiences); intertextuality (e.g., information that a reader or viewer brings to a text from other texts experienced previously); gaps in the text (e.g., information that is left out and that the reader or viewer must fill in); and silences in the text (e.g., the absence of the voices of certain people or groups).

Students who are critically literate are able, for example, to actively analyse media messages and determine possible motives and underlying messages. They are able to determine what biases might be contained in texts, media, and resource material and why that might be, how the content of these materials might be determined and by whom, and whose perspectives might have been left out and why. Only then are students equipped to produce their own interpretation of an issue. Opportunities should be provided for students to engage in a critical discussion of “texts”, including books and textbooks, television programs, movies, documentaries, web pages, advertising, music, gestures, oral texts, newspaper and magazine articles, letters, cultural text forms, stories, and other forms of expression. Such discussions empower students to understand the impact on members of society that was intended by the text’s creators. Language and communication are never neutral: they are used to inform, entertain, persuade, and manipulate.

The literacy skill of *metacognition* supports students’ ability to think critically through reflection on their own thought processes. Acquiring and using metacognitive skills has emerged as a powerful approach for promoting a focus on thinking skills in literacy and across all disciplines, and for empowering students with the skills needed to monitor their own learning. As they reflect on their strengths and needs, students are encouraged to advocate for themselves to get the support they need in order to achieve their goals.

Mathematical Literacy

[View in ASL](#)



Mathematical literacy is an individual’s capacity to formulate, employ, and interpret mathematics in a variety of contexts. It includes reasoning mathematically and using mathematical concepts, procedures, facts, and tools to describe, explain, and predict phenomena. It assists individuals to recognize the role that mathematics plays in the world and to make the well-founded judgments and decisions needed by constructive, engaged, and reflective citizens.

- Council of Ministers of Education, Canada (CMEC), [Measuring Up: Canadian Results of the OECD PISA Study](#), 2016, p. 10

The Importance of Mathematical Literacy¹⁰

Mathematical literacy involves more than executing procedures. It implies a knowledge base and the competence and confidence to apply this knowledge in the practical world. A mathematically literate

¹⁰ Adapted from *Leading Math Success: Mathematical Literacy, Grades 7–12 – The Report of the Expert Panel on Student Success in Ontario*, 2004, pages 10 and 24.

person can estimate; interpret data; solve day-to-day problems; reason in numerical, graphical, and geometric situations; and communicate using mathematics.

As knowledge expands and the economy evolves, more people are working with technologies or working in settings where mathematics is a cornerstone. Problem solving, the processing of information, and communication are becoming routine job requirements. Outside the workplace, mathematics arises in many everyday situations. Mathematical literacy is necessary both at work and in daily life.

Mathematical literacy is as important as proficiency in reading and writing. Mathematics is so entwined with today's way of life that we cannot fully comprehend the information that surrounds us without a basic understanding of mathematical ideas. Confidence and competence in mathematics lead to productive participation in today's complex information society, and open the door to opportunity.

The Scope of Mathematical Literacy

Mathematical literacy encompasses the ability to:

- estimate in numerical or geometric situations
- know and understand mathematical concepts and procedures
- question, reason, and solve problems
- make connections within mathematics and between mathematics and life
- generate, interpret, and compare data
- communicate mathematical reasoning

Mathematical literacy has several dimensions – for example, numerical literacy, spatial literacy, and data literacy – and extends beyond the mathematics classroom to other fields of study.

Teachers should take advantage of the abundant opportunities that exist for fostering mathematical literacy across the curriculum. All teachers have a responsibility to communicate the view that all students can and should do mathematics.

Environmental Education

[View in ASL](#)

Environmental education is both the responsibility of the entire education community and a rich opportunity for cross-curricular learning. It can be taught across subjects and grades, providing context that can enrich and enliven learning in all subject areas. It also provides opportunities for critical thinking, learning about citizenship, and developing personal responsibility. It offers students the opportunity to develop a deeper understanding of themselves, their role in society, and their dependence on one another and on the Earth's natural systems.

The curriculum provides opportunities for students to learn about environmental processes, issues, and solutions, and to demonstrate their learning as they practise and promote environmental stewardship at school and in their communities.

[*Acting Today, Shaping Tomorrow: A Policy Framework for Environmental Education in Ontario Schools*](#) outlines an approach to environmental education that recognizes the need for all Ontario students to learn “in, about and/or for” the environment, and promotes environmental responsibility on the part of students, school staff, and leaders at all levels of the education system.

Resource documents – [*The Ontario Curriculum, Grades 1–8 and The Kindergarten Program: Environmental Education, Scope and Sequence of Expectations, 2017*](#) and [*The Ontario Curriculum, Grades 9–12: Environmental Education, Scope and Sequence of Expectations, 2017*](#) – have been prepared to assist teachers in planning lessons that integrate environmental education with other subject areas. They identify curriculum expectations and related examples and prompts in disciplines across the Ontario curriculum that provide opportunities for student learning “in, about, and/or for” the environment. Teachers can use these documents to plan lessons that relate explicitly to the environment, or they can draw on them for opportunities to use the environment as the *context for learning*. These documents can also be used to make curriculum connections to school-wide environmental initiatives.

Social-Emotional Learning Skills

[View in ASL](#)

The development of social-emotional learning (SEL) skills helps students foster overall health and well-being, positive mental health, and the ability to learn, build resilience, and thrive.

Students will learn skills to:	So that they can:
• identify and manage emotions	• express their feelings and understand the feelings of others
• recognize sources of stress and cope with challenges	• develop personal resilience
• maintain positive motivation and perseverance	• foster a sense of optimism and hope
• build relationships and communicate effectively	• support healthy relationships and respect diversity
• develop self-awareness and self-confidence	• develop a sense of identity and belonging
• think critically and creatively	• make informed decisions and solve problems

Social-emotional learning skills are an explicit component of learning in the elementary health and physical education curriculum. However, there are opportunities for students to develop SEL skills in

connection with their learning in all subjects and disciplines. Skills to support mental health and well-being can be developed across the curriculum, in the context of school activities, at home, and in the community.

It is beneficial for students to make connections between SEL skills, [transferable skills](#), and learning skills and work habits (see [Growing Success](#), 2010, Chapter 2). Taken together, these interrelated skills support students' overall health and well-being, positive mental health, and the ability to learn and to become lifelong learners. They enhance students' experience in school and beyond, preparing them to succeed personally and to become economically productive and actively engaged citizens. [School Mental Health Ontario](#) (SMHO) has resources to support the development of social-emotional learning in Ontario schools.

Transferable skills

Introduction

[View in ASL](#)

The Importance of Transferable Skills in the Curriculum

Today's graduates will enter a world that is more competitive, more globally connected, and more technologically engaged than it has been in any other period of history. Over the course of the next decade, millions of young Canadians will enter a workforce that is dramatically different from the one we know today. With the growing automation of jobs, extraordinary technological advancements, and the realities of a global economy, students will need to be prepared for job flexibility, frequent career re-orientation, and work and civic life in a globalized, digital age. Equipping students with transferable skills and a desire for lifelong learning will help to prepare them for these new realities, and to navigate and shape their future successfully.

Transferable skills are the skills and attributes that students need in order to thrive in the modern world. Based on international research, information provided by employers, and its work with jurisdictions across Canada, the Ontario Ministry of Education has defined seven important categories of transferable skills – sometimes referred to as “competencies”¹¹ – that will help students navigate the world of work and meet with success in the future:

¹¹ These categories of transferable skills are aligned with the [six “global competencies”](#) developed collaboratively by ministers of education across Canada on the basis of the competencies outlined in *21st Century Competencies: Foundation Document for Discussion* (Ontario Ministry of Education, 2016). The global competencies were then published by the Council of Ministers of Education, Canada (CMEC) as part of an effort to prepare students across the nation for a complex and unpredictable future with

- critical thinking and problem solving
- innovation, creativity, and entrepreneurship
- self-directed learning
- collaboration
- communication
- global citizenship and sustainability
- digital literacy

These seven broad categories of skills, necessary in today’s rapidly changing world, can be seen as a framework encompassing the wide range of discrete transferable skills that students acquire over time. Developing transferable skills essentially means “learning for transfer” – that is, taking what is learned in one situation and applying it to other, new situations. Students in Ontario schools “learn for transfer” in all of the subjects and disciplines of the Ontario curriculum, from Kindergarten to Grade 12. In fact, in every grade and subject, their learning is assessed, in part, in terms of their ability to apply or transfer what they have learned to familiar and new contexts (see the category “Application” in the [Sample Achievement Charts](#)). The curriculum provides opportunities for students to develop transferable skills in age- and grade-appropriate ways throughout their school years. Students develop transferable skills not in isolation but as part of their learning in all subjects of the curriculum. These skills are developed through students’ cognitive, social, emotional, and physical engagement in learning. Educators facilitate students’ development of transferable skills explicitly through a variety of teaching and learning methods, models, and approaches, and assessment practices, in a safe, inclusive, and equitable learning environment.

Critical Thinking and Problem Solving

[View in ASL](#)

Definition

Critical thinking and problem solving involve locating, processing, analysing, and interpreting relevant and reliable information to address complex issues and problems, make informed judgements and decisions, and take effective action. With critical thinking skills comes an awareness that solving problems can have a positive impact in the world, and this contributes to achieving one’s potential as a constructive and reflective citizen. Learning is deepened when it occurs in the context of authentic and meaningful real-world experiences.

rapidly changing political, social, economic, technological, and environmental landscapes. The new categories of transferable skills outlined here have been updated on the basis of current research, and a seventh category – “digital literacy” – has been added.

Student Descriptors

- Students engage in inquiry processes that include locating, processing, interpreting, synthesizing, and critically analysing information in order to solve problems and make informed decisions. These processes involve critical, digital, and data literacy.
- Students solve meaningful and complex real-life problems by taking concrete steps – identifying and analysing the problem, creating a plan, prioritizing actions to be taken, and acting on the plan – as they address issues and design and manage projects.
- Students detect patterns, make connections, and transfer or apply what they have learned in a given situation to other situations, including real-world situations.
- Students construct knowledge and apply what they learn to all areas of their lives – at school, home, and work; among friends; and in the community – with a focus on making connections and understanding relationships.
- Students analyse social, economic, and ecological systems to understand how they function and how they interrelate.

Innovation, Creativity, and Entrepreneurship

[View in ASL](#)

Definition

Innovation, creativity, and entrepreneurship support the ability to turn ideas into action in order to meet the needs of a community. These skills include the capacity to develop concepts, ideas, or products for the purpose of contributing innovative solutions to economic, social, and environmental problems. Developing these skills involves a willingness to assume leadership roles, take risks, and engage in independent, unconventional thinking in the context of experimenting, conducting research, and exploring new strategies, techniques, and perspectives. An entrepreneurial mindset understands the importance of building and scaling ideas for sustainable growth.

Student Descriptors

- Students formulate and express insightful questions and opinions to generate novel ideas.
- Students contribute solutions to economic, social, and environmental problems in order to meet a need in a community by: enhancing concepts, ideas, or products through a creative process; taking risks in their creative thinking as they devise solutions; making discoveries through inquiry research, by testing hypotheses and experimenting with new strategies or techniques.

- Students demonstrate leadership, initiative, imagination, creativity, spontaneity, and ingenuity as they engage in a range of creative processes, motivating others with their ethical entrepreneurial spirit.

Self-Directed Learning

[View in ASL](#)

Definition

Self-directed learning involves becoming aware of and managing one's own process of learning. It includes developing dispositions that support motivation, self-regulation, perseverance, adaptability, and resilience. It also calls for a growth mindset – a belief in one's ability to learn – combined with the use of strategies for planning, reflecting on, and monitoring progress towards one's goals, and reviewing potential next steps, strategies, and results. Self-reflection and thinking about thinking (metacognition) support lifelong learning, adaptive capacity, well-being, and the ability to transfer learning in an ever-changing world.

Student Descriptors

- Students learn to think about their own thinking and learning (metacognition) and to believe in their ability to learn and grow (growth mindset). They develop their ability to set goals, stay motivated, and work independently.
- Students who regulate their own learning are better prepared to become lifelong learners. They reflect on their thinking, experiences, and values, and respond to critical feedback, to enhance their learning. They also monitor the progress of their learning.
- Students develop a sense of identity in the context of Canada's various and diverse communities.
- Students cultivate emotional intelligence to better understand themselves and others and build healthy relationships.
- Students learn to take the past into account in order to understand the present and approach the future in a more informed way.
- Students develop personal, educational, and career goals and persevere to overcome challenges in order to reach those goals. They learn to adapt to change and become resilient in the face of adversity.
- Students become managers of the various aspects of their lives – cognitive, emotional, social, physical, and spiritual – to enhance their mental health and overall well-being.

Collaboration

[View in ASL](#)

Definition

Collaboration involves the interplay of the cognitive (thinking and reasoning), interpersonal, and intrapersonal competencies needed to work with others effectively and ethically. These skills deepen as they are applied, with increasing versatility, to co-construct knowledge, meaning, and content with others in diverse situations, both physical and virtual, that involve a variety of roles, groups, and perspectives.

Student Descriptors

- Students participate successfully in teams by building positive and respectful relationships, developing trust, and acting cooperatively and with integrity.
- Students learn from others and contribute to their learning as they co-construct knowledge, meaning, and content.
- Students assume various roles on the team, respect a diversity of perspectives, and recognize different sources of knowledge, including Indigenous ways of knowing.
- Students address disagreements and manage conflict in a sensitive and constructive manner.
- Students interact with a variety of communities and/or groups and use various technologies appropriately to facilitate working with others.

Communication

[View in ASL](#)

Definition

Communication involves receiving and expressing meaning (e.g., through reading and writing, viewing and creating, listening and speaking) in different contexts and with different audiences and purposes. Effective communication increasingly involves understanding local and global perspectives and societal and cultural contexts, and using a variety of media appropriately, responsibly, safely, and with a view to creating a positive digital footprint.

Student Descriptors

- Students communicate effectively in different contexts, orally and in writing, using a variety of media.

- Students communicate using the appropriate digital tools, taking care to create a positive digital footprint.
- Students ask effective questions to acquire knowledge; listen to all points of view and ensure that those views are heard; voice their own opinions; and advocate for ideas.
- Students learn about a variety of languages, including Indigenous languages, and understand the cultural importance of language.

Global Citizenship and Sustainability

[View in ASL](#)

Definition

Global citizenship and sustainability involves understanding diverse world views and perspectives in order to effectively address the various political, environmental, social, and economic issues that are central to living sustainably in today's interconnected and interdependent world. It also involves acquiring the knowledge, motivation, dispositions, and skills required for engaged citizenship, along with an appreciation of the diversity of people and perspectives in the world. It calls for the ability to envision and work towards a better and more sustainable future for all.

Student Descriptors

- Students understand the political, environmental, economic, and social forces at play in the world today, how they interconnect, and how they affect individuals, communities, and countries.
- Students make responsible decisions and take actions that support quality of life for all, now and in the future.
- Students recognize discrimination and promote principles of equity, human rights, and democratic participation.
- Students recognize the traditions, knowledge, and histories of Indigenous peoples, appreciate their historical and contemporary contributions to Canada, and recognize the legacy of residential schools.
- Students learn from and with people of diverse cultures and backgrounds and develop cross-cultural understanding.
- Students engage in local, national, and global initiatives to make a positive difference in the world.
- Students contribute to society and to the culture of local, national, and global communities, both physical and virtual, in a responsible, inclusive, sustainable, ethical, and accountable manner.

- Students, as citizens, participate in various groups and online networks in a safe and socially responsible manner.

Digital Literacy

[View in ASL](#)

Definition

Digital literacy involves the ability to solve problems using technology in a safe, legal, and ethically responsible manner. With the ever-expanding role of digitalization and big data in the modern world, digital literacy also means having strong data literacy skills and the ability to engage with emerging technologies. Digitally literate students recognize the rights and responsibilities, as well as the opportunities, that come with living, learning, and working in an interconnected digital world.

Student Descriptors

- Students select and use appropriate digital tools to collaborate, communicate, create, innovate, and solve problems.
- Students understand how to manage and regulate their use of technology to support their mental health and well-being.
- Students use digital tools to define and plan data searches, collect data, and identify relevant data sets. They analyse, interpret, and graphically represent, or “visualize”, data in various ways to solve problems and inform decisions.
- Students demonstrate a willingness and confidence to explore and use new or unfamiliar digital tools and emerging technologies (e.g., open source software, wikis, robotics, augmented reality). Students understand how different technologies are connected and recognize their benefits and limitations.
- Students manage their digital footprint by engaging in social media and online communities respectfully, inclusively, safely, legally, and ethically. Students understand their rights with respect to personal data and know how to protect their privacy and security and respect the privacy and security of others.
- Students analyse and understand the impact of technological advancements on society, and society’s role in the evolution of technology.

Assessment and Evaluation

Introduction

[View in ASL](#)

[*Growing Success: Assessment, Evaluation, and Reporting in Ontario Schools, First Edition, Covering Grades 1 to 12, 2010*](#) sets out the Ministry of Education's assessment, evaluation, and reporting policy. The policy aims to maintain high standards, improve student learning, and benefit students, parents¹², and teachers in elementary and secondary schools across the province. Successful implementation of this policy depends on the professional judgement¹³ of educators at all levels as well as on their ability to work together and to build trust and confidence among parents and students.

A brief summary of some major aspects of the current assessment, evaluation, and reporting policy is given below. Teachers should refer to *Growing Success* for more detailed information.

Fundamental Principles

[View in ASL](#)

The primary purpose of assessment and evaluation is to improve student learning.

The seven fundamental principles given below (excerpted from *Growing Success*, page 6) lay the foundation for rich and challenging practice. When these principles are fully understood and observed by all teachers, they will guide the collection of meaningful information that will help inform instructional decisions, promote student engagement, and improve student learning.

To ensure that assessment, evaluation, and reporting are valid and reliable, and that they lead to the improvement of learning for all students, teachers use practices and procedures that:

- are fair, transparent, and equitable;
- support all students, including those with special education needs, those who are learning the language of instruction (English or French), and those who are First Nation, Métis, or Inuit;

¹² The word *parent(s)* is used on this website to refer to parent(s) and guardian(s). It may also be taken to include caregivers or close family members who are responsible for raising the child.

¹³ "Professional judgement", as defined in [Growing Success \(p. 152\)](#), is "judgement that is informed by professional knowledge of curriculum expectations, context, evidence of learning, methods of instruction and assessment, and the criteria and standards that indicate success in student learning. In professional practice, judgement involves a purposeful and systematic thinking process that evolves in terms of accuracy and insight with ongoing reflection and self-correction".

- are carefully planned to relate to the curriculum expectations and learning goals and, as much as possible, to the interests, learning styles and preferences, needs, and experiences of all students;
- are communicated clearly to students and parents at the beginning of the school year or course and at other appropriate points throughout the school year or course;
- are ongoing, varied in nature, and administered over a period of time to provide multiple opportunities for students to demonstrate the full range of their learning;
- provide ongoing descriptive feedback that is clear, specific, meaningful, and timely to support improved learning and achievement;
- develop students' self-assessment skills to enable them to assess their own learning, set specific goals, and plan next steps for their learning.

Culturally Responsive and Relevant Assessment and Evaluation

[Culturally responsive and relevant pedagogy \(CRRP\)](#) reflects and affirms students' cultural and social identities, languages, and family structures. It involves careful acknowledgement, respect, and understanding of the similarities and differences among students, and between students and teachers, in order to respond effectively to student thinking and promote student learning.

Engaging in assessment from a CRRP stance requires that teachers gain awareness of and reflect on their own beliefs about who a learner is and what they can achieve (see the questions for consideration provided below). In this process, teachers engage in continual self-reflection – and the critical analysis of various data – to understand and address the ways in which teacher identity and bias affect the assessment and evaluation of student learning. Assessment from a CRRP stance starts with having a deep knowledge of every student and an understanding of how they learn best.

The primary purpose of assessment is to improve student learning. Assessment *for* learning creates opportunities for teachers to intentionally learn about each student and their sociocultural and linguistic background in order to gather a variety of evidence about their learning in a way that is reflective of and responsive to each student's strengths, experiences, interests, and cultural ways of knowing. Ongoing descriptive feedback and responsive coaching are essential for improving student learning.

Teachers engage in assessment *as* learning by creating ongoing opportunities for all students to develop their capacity to be confident, independent, autonomous learners who set individual goals, monitor their own progress, determine next steps, and reflect on their thinking and learning in relation to learning goals and curriculum expectations. One way in which teachers differentiate assessment is by providing tasks that allow multiple entry points for all students and that enable all students to design and create personally meaningful assignments, projects, performances, and other demonstrations of their learning.

Assessment *of* learning is used by the teacher to summarize student learning at a given point in time. This summary is used to make judgements about the quality of student learning on the basis of established criteria, to assign a value to represent that quality, and to support the communication of information about achievement to each student and to parents, teachers, and others.

The evidence that is collected about student learning, including through observations and conversations as well as student products, should reflect and affirm the student's lived experiences within their school, home, and community, their learning strengths, and their knowledge of concepts and skills. This process of triangulating evidence of student learning allows teachers to improve their understanding of how each student is progressing in their learning.

When teachers engage in the process of examining their own biases regarding classroom assessment and evaluation practices, they might consider some of the following questions:

- Are the tasks accessible to, and inclusive of, all learners? Do the tasks include appropriate and varied entry points for all students?
- Do the tasks connect to students' prior learning and give them opportunities to be sense makers and to integrate their new learning? Do the selected tasks reflect students' identities and lived experiences?
- Do all students have equitable access to the tools they need to complete the tasks being set?
- What opportunities can teachers build into their practice to offer students descriptive feedback to enhance learning? Are graded assessment tasks used in a way that complements the use of descriptive feedback for growth?
- How can information be conveyed about students' learning progress to students and parents in an ongoing and meaningful way?
- What is the purpose of assigning and grading a specific task or activity? Are student choice and agency considered?
- How do teacher biases influence decisions about what tasks or activities are chosen for assessment?

Learning Skills and Work Habits

[View in ASL](#)

The development of learning skills and work habits is an integral part of a student's learning. To the extent possible, however, the evaluation of learning skills and work habits, apart from any that may be included as part of a curriculum expectation in a course, should *not* be considered in the determination of a student's grades. Assessing, evaluating, and reporting on the achievement of curriculum expectations and on the demonstration of learning skills and work habits *separately* allows teachers to provide information to the parents and student that is specific to each of these two areas.

The six learning skills and work habits are responsibility, organization, independent work, collaboration, initiative, and self-regulation.

Content Standards and Performance Standards

[View in ASL](#)

The Ontario curriculum for Grades 1 to 12 comprises *content standards* and *performance standards*. Assessment and evaluation will be based on both the content standards and the performance standards.

The content standards are the overall and specific curriculum expectations given in the curriculum for every subject and discipline.

The performance standards are outlined in the achievement chart, also provided in the curriculum for every subject and discipline (each achievement chart is specific to the subject/discipline; see the [sample charts provided](#)). The achievement chart is a standard province-wide guide and is to be used by all teachers as a framework for assessing and evaluating student achievement of the expectations in the particular subject or discipline. It enables teachers to make consistent judgements about the quality of student learning, based on clear performance standards and on a body of evidence collected over time. It also provides teachers with a foundation for developing clear and specific feedback for students and parents.

The purposes of the achievement chart are to:

- provide a common framework that encompasses all curriculum expectations for all subjects/courses across the grades;
- guide the development of high-quality assessment tasks and tools (including rubrics);
- help teachers plan instruction for learning;
- provide a basis for consistent and meaningful feedback to students in relation to provincial content and performance standards;
- establish categories and criteria for assessing and evaluating students' learning.

Assessment “for Learning” and “as Learning”

[View in ASL](#)

Assessment is the process of gathering information that accurately reflects how well a student is achieving the curriculum expectations in a grade or course. The primary purpose of assessment is to improve student learning. Assessment for the purpose of improving student learning is seen as both “assessment *for* learning” and “assessment *as* learning”. As part of assessment *for* learning, teachers provide students with descriptive feedback and coaching for improvement. Teachers engage in

assessment *as* learning by helping all students develop their capacity to be independent, autonomous learners who are able to set individual goals, monitor their own progress, determine next steps, and reflect on their thinking and learning.

As essential steps in assessment *for* learning and *as* learning, teachers need to:

- plan assessment concurrently and integrate it seamlessly with instruction;
- share learning goals and success criteria with students at the outset of learning to ensure that students and teachers have a common and shared understanding of these goals and criteria as learning progresses;
- gather information about student learning before, during, and at or near the end of a period of instruction, using a variety of assessment strategies and tools;
- use assessment to inform instruction, guide next steps, and help students monitor their progress towards achieving their learning goals;
- analyse and interpret evidence of learning;
- give and receive specific and timely descriptive feedback about student learning;
- help students to develop skills of peer assessment and self-assessment.

Evaluation

[View in ASL](#)

Evaluation refers to the process of judging the quality of student learning on the basis of established performance standards, and assigning a value to represent that quality. Evaluation accurately summarizes and communicates to parents, other teachers, employers, institutions of further education, and students themselves what students know and can do with respect to the overall curriculum expectations. Evaluation is based on assessment *of* learning that provides evidence of student achievement at strategic times throughout the course, often at the end of a period of learning.

All curriculum expectations must be accounted for in instruction and assessment, but *evaluation focuses on students' achievement of the overall expectations*¹⁴. Each student's achievement of the overall expectations is evaluated on the basis of the student's achievement of related specific expectations. The overall expectations are broad in nature, and the specific expectations define the particular content or scope of the knowledge and skills referred to in the overall expectations. Teachers will use their professional judgement to determine which specific expectations should be used to evaluate

¹⁴ Beginning in the 2021–22 school year, schools are asked not to assess, evaluate or report on the overall expectations related to social-emotional learning skills in *The Ontario Curriculum, Grades 1–8, Mathematics (2020)* and *The Ontario Curriculum, Grades 1–8, Health and Physical Education (2019)*. It is the ministry's expectation that instruction of the social-emotional learning skills will continue while educators engage in ongoing professional learning.

achievement of the overall expectations, and which ones will be accounted for in instruction and assessment but not necessarily evaluated.

Determining a report card grade involves the interpretation of evidence collected through observations, conversations, and student products (tests/exams, assignments for evaluation), combined with the teacher's professional judgement and consideration of factors such as the number of tests/exams or assignments for evaluation that were not completed or submitted and the fact that some evidence may carry greater weight than other evidence.

Secondary

Seventy per cent of the final grade (a percentage mark) in a course will be based on evaluation conducted throughout the course. This portion of the grade should reflect the student's most consistent level of achievement, with special consideration given to more recent evidence. Thirty per cent will be based on a final evaluation administered at or towards the end of the course. This evaluation will be based on evidence from one or a combination of the following: an examination, a performance, an essay, and/or another method of evaluation suitable to the course content. The final evaluation allows the student an opportunity to demonstrate comprehensive achievement of the overall expectations for the course.

Reporting Student Achievement

[View in ASL](#)

Elementary

Three formal report cards are issued in Ontario's publicly funded elementary schools, as described below.

The Elementary Progress Report Card shows a student's development of learning skills and work habits during the fall of the school year, as well as the student's general progress in working towards achievement of the curriculum expectations in each subject (reported as "progressing very well", "progressing well", or "progressing with difficulty").

The Elementary Provincial Report Card shows a student's achievement at specific points in the school year. The first Provincial Report Card reflects student achievement of the overall curriculum expectations introduced and developed from September to January/February of the school year, as well as the student's development of learning skills and work habits during that period. The second reflects achievement of curriculum expectations introduced or further developed from January/February to June, as well as further development of learning skills and work habits during that period. The Provincial Report Card for Grades 1–6 uses letter grades; the report card for Grades 7 and 8 uses percentage grades.

Secondary

The Provincial Report Card, Grades 9–12, shows a student’s achievement at specific points in the school year or semester. There are two formal reporting periods for a semestered course and three formal reporting periods for a non-semestered course. The reports reflect student achievement of the overall curriculum expectations, as well as development of learning skills and work habits.

Communication with parents and students

Although there are formal reporting periods, communication with parents and students about student achievement should be continuous throughout the year or course, by a variety of means, such as parent-teacher or parent-student-teacher conferences, portfolios of student work, student-led conferences, interviews, phone calls, checklists, and informal reports. Communication about student achievement should be designed to provide detailed information that will encourage students to set goals for learning, help teachers to establish plans for teaching, and assist parents in supporting learning at home.

Categories of Knowledge and Skills

[View in ASL](#)

The categories represent four broad areas of knowledge and skills within which the expectations for any given subject or course can be organized. The four categories should be considered as interrelated, reflecting the wholeness and interconnectedness of learning.

The categories help teachers focus not only on students’ acquisition of knowledge but also on their development of the skills of thinking, communication, and application.

The categories of knowledge and skills are as follows:

Knowledge and Understanding. Subject-specific content acquired in each grade or course (knowledge), and the comprehension of its meaning and significance (understanding).

Thinking. The use of critical and creative thinking skills and/or processes.

Communication. The conveying of meaning and expression through various forms.

Application. The use of knowledge and skills to make connections within and between various contexts.

In all subjects and courses, students should be given numerous and varied opportunities to demonstrate the full extent of their achievement of the curriculum expectations across all four categories of knowledge and skills.

Teachers will ensure that student learning is assessed and evaluated in a balanced manner with respect to the four categories, and that achievement of particular expectations is considered within the appropriate categories. The emphasis on “balance” reflects the fact that all categories of the achievement chart are important and need to be a part of the process of instruction, learning, assessment, and evaluation. However, it also indicates that for different courses, the *relative* importance of each of the categories may vary. The importance accorded to each of the four categories in assessment and evaluation should reflect the emphasis accorded to them in the curriculum expectations for the subject or course and in instructional practice.

Criteria and Descriptors

[View in ASL](#)

To further guide teachers in their assessment and evaluation of student learning, the achievement chart provides “criteria” and “descriptors”.

A set of criteria is identified for each category in the achievement chart. The criteria are subsets of the knowledge and skills that define the category. The criteria identify the aspects of student performance that are assessed and/or evaluated, and they serve as a guide to what teachers look for. Each curriculum has subject- or discipline-specific criteria and descriptors. For example, in the English curriculum, in the Knowledge and Understanding category, the criteria are “knowledge of content” and “understanding of content”. The former includes examples such as forms of text and elements of style, and the latter includes examples such as relationships among facts. “Descriptors” indicate the characteristics of the student’s performance, with respect to a particular criterion, on which assessment or evaluation is focused. *Effectiveness* is the descriptor used for each of the criteria in the Thinking, Communication, and Application categories. What constitutes effectiveness in any given performance task will vary with the particular criterion being considered. Assessment of effectiveness may therefore focus on a quality such as appropriateness, clarity, accuracy, precision, logic, relevance, significance, fluency, flexibility, depth, or breadth, as appropriate for the particular criterion.

Levels of Achievement

[View in ASL](#)

The achievement chart also identifies four levels of achievement, defined as follows:

Level 1 represents achievement that falls much below the provincial standard. The student demonstrates the specified knowledge and skills with limited effectiveness. Students must work at significantly improving in specific areas, as necessary, if they are to be successful in a subject or course in the next grade.

Level 2 represents achievement that approaches the standard. The student demonstrates the specified knowledge and skills with some effectiveness. Students performing at this level need to work on identified learning gaps to ensure future success.

Level 3 represents the provincial standard for achievement. The student demonstrates the specified knowledge and skills with considerable effectiveness. Parents of students achieving at level 3 can be confident that their children will be prepared for work in subsequent grades or courses.

Level 4 identifies achievement that surpasses the provincial standard. The student demonstrates the specified knowledge and skills with a high degree of effectiveness. *However, achievement at level 4 does not mean that the student has achieved expectations beyond those specified for the grade or course.*

Specific “qualifiers” are used with the descriptors in the achievement chart to describe student performance at each of the four levels of achievement – the qualifier *limited* is used for level 1; *some* for level 2; *considerable* for level 3; and *a high degree of or thorough* for level 4. Hence, achievement at level 3 in the Thinking category for the criterion “use of planning skills” would be described in the achievement chart as “[The student] uses planning skills with *considerable* effectiveness”.

Sample Achievement Charts

[View in ASL](#)

Three samples of the achievement chart are provided, from the following subjects/disciplines:

- The Arts, Grades 1–8
- Science and Technology, Grades 1–8
- English, Grades 11 and 12

These three samples illustrate the consistent characteristics of the performance standards across all subjects and disciplines and across all grades. The samples also illustrate how the achievement chart varies – particularly with respect to the examples provided for the criteria in each category – to reflect the nature of the particular subject or discipline. For instance, the examples for the criterion “Application of knowledge and skills” in the Application category of the achievement chart for the arts include performance skills, composition, and choreography, whereas those for science and technology include investigation skills and safe use of equipment and technology.

As discussed in the preceding sections, the achievement chart identifies four categories of knowledge and skills and four levels of achievement in the particular subject/discipline.

The Achievement Chart for The Arts, Grades 1–8

Knowledge and Understanding – Subject-specific content acquired in each grade (knowledge), and the comprehension of its meaning and significance (understanding)

Categories	Level 1	Level 2	Level 3	Level 4
Knowledge of content (e.g., facts, genres, terms, definitions, techniques, elements, principles, forms, structures, conventions)	The student demonstrates limited knowledge of content.	The student demonstrates some knowledge of content.	The student demonstrates considerable knowledge of content.	The student demonstrates thorough knowledge of content.
Understanding of content (e.g., concepts, ideas, procedures, processes, themes, relationships among elements, informed opinions)	The student demonstrates limited understanding of content.	The student demonstrates some understanding of content.	The student demonstrates considerable understanding of content.	The student demonstrates thorough understanding of content.

Thinking – The use of critical and creative thinking skills and/or processes

Categories	Level 1	Level 2	Level 3	Level 4
Use of planning skills (e.g., formulating questions, generating ideas, gathering information, focusing research, outlining, organizing an arts presentation or project, brainstorming/ bodystorming, blocking, sketching, using visual organizers, listing goals in a rehearsal log, inventing notation)	The student uses planning skills with limited effectiveness.	The student uses planning skills with some effectiveness.	The student uses planning skills with considerable effectiveness.	The student uses planning skills with a high degree of effectiveness.
Use of processing skills (e.g., analysing, evaluating, inferring, interpreting, editing, revising, refining, forming conclusions, detecting bias, synthesizing)	The student uses processing skills with limited effectiveness.	The student uses processing skills with some effectiveness.	The student uses processing skills with considerable effectiveness.	The student uses processing skills with a high degree of effectiveness.
Use of critical/creative thinking processes (e.g., creative and analytical processes, design process, exploration of the elements, problem solving, reflection, elaboration, oral discourse, evaluation, critical literacy, metacognition, invention, critiquing, reviewing)	The student uses critical/creative thinking processes with limited effectiveness.	The student uses critical/creative thinking processes with some effectiveness.	The student uses critical/creative thinking processes with considerable effectiveness.	The student uses critical/creative thinking processes with a high degree of effectiveness.

Communication – The conveying of meaning through various forms

Categories	Level 1	Level 2	Level 3	Level 4
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Expression and organization of ideas and understandings in art forms (dance, drama, music, and the visual arts), including media/ multimedia forms (e.g., expression of ideas and feelings using visuals, movements, the voice, gestures, phrasing, techniques), and in oral and written forms (e.g., clear expression and logical organization in critical responses to art works and informed opinion pieces)	The student expresses and organizes ideas and understandings with limited effectiveness.	The student expresses and organizes ideas and understandings with some effectiveness.	The student expresses and organizes ideas and understandings with considerable effectiveness.	The student expresses and organizes ideas and understandings with a high degree of effectiveness.
Communication for different audiences (e.g., peers, adults, younger children) and purposes through the arts (e.g., drama presentations, visual arts exhibitions, dance and music performances) and in oral and written forms (e.g., debates, analyses)	The student communicates for different audiences and purposes with limited effectiveness.	The student communicates for different audiences and purposes with some effectiveness.	The student communicates for different audiences and purposes with considerable effectiveness.	The student communicates for different audiences and purposes with a high degree of effectiveness.
Use of conventions in dance, drama, music, and the visual arts (e.g., allegory, narrative or symbolic representation, style, articulation, drama conventions, choreographic forms, movement vocabulary) and arts vocabulary and terminology in oral and written forms	The student uses conventions, vocabulary, and terminology of the arts with limited effectiveness.	The student uses conventions, vocabulary, and terminology of the arts with some effectiveness.	The student uses conventions, vocabulary, and terminology of the arts with considerable effectiveness.	The student uses conventions, vocabulary, and terminology of the arts with a high degree of effectiveness.

Application – The use of knowledge and skills to make connections within and between various contexts

Categories	Level 1	Level 2	Level 3	Level 4
Application of knowledge and skills (e.g., performance skills, composition, choreography, elements, principles, processes, technologies, techniques, strategies, conventions) in familiar contexts (e.g., guided improvisation, performance of a familiar work, use of familiar forms)	The student applies knowledge and skills in familiar contexts with limited effectiveness.	The student applies knowledge and skills in familiar contexts with some effectiveness.	The student applies knowledge and skills in familiar contexts with considerable effectiveness.	The student applies knowledge and skills in familiar contexts with a high degree of effectiveness.

Transfer of knowledge and skills (e.g., concepts, strategies, processes, techniques) to new contexts (e.g., a work requiring stylistic variation, an original composition, student-led choreography, an interdisciplinary or multidisciplinary project)	The student transfers knowledge and skills to new contexts with limited effectiveness.	The student transfers knowledge and skills to new contexts with some effectiveness.	The student transfers knowledge and skills to new contexts with considerable effectiveness.	The student transfers knowledge and skills to new contexts with a high degree of effectiveness.
Making connections within and between various contexts (e.g., between the arts; between the arts and personal experiences and the world outside the school; between cultural and historical, global, social, and/or environmental contexts; between the arts and other subjects)	The student makes connections within and between various contexts with limited effectiveness.	The student makes connections within and between various contexts with some effectiveness.	The student makes connections within and between various contexts with considerable effectiveness.	The student makes connections within and between various contexts with a high degree of effectiveness.

The Achievement Chart for Science and Technology, Grades 1–8

Knowledge and Understanding – Subject-specific content acquired in each grade (knowledge), and the comprehension of its meaning and significance (understanding)

Categories	Level 1	Level 2	Level 3	Level 4
Knowledge of content (e.g., facts, terminology, definitions)	The student demonstrates limited knowledge of content.	The student demonstrates some knowledge of content.	The student demonstrates considerable knowledge of content.	The student demonstrates thorough knowledge of content.
Understanding of content (e.g., concepts, ideas, theories, principles, procedures, processes)	The student demonstrates limited understanding of content.	The student demonstrates some understanding of content.	The student demonstrates considerable understanding of content.	The student demonstrates thorough understanding of content.

Thinking and Investigation – The use of critical and creative thinking skills and inquiry and problem-solving skills and/or processes

Categories	Level 1	Level 2	Level 3	Level 4
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Use of initiating and planning skills and strategies (e.g., <i>formulating questions, identifying the problem, developing hypotheses, scheduling, selecting strategies and resources, developing plans</i>)	The student uses initiating and planning skills and strategies with limited effectiveness.	The student uses initiating and planning skills and strategies with some effectiveness.	The student uses initiating and planning skills and strategies with considerable effectiveness.	The student uses initiating and planning skills and strategies with a high degree of effectiveness.
Use of processing skills and strategies (e.g., <i>performing and recording; gathering evidence and data; examining different points of view; selecting tools, equipment, materials, and technology; observing; manipulating materials; proving</i>)	The student uses processing skills and strategies with limited effectiveness.	The student uses processing skills and strategies with some effectiveness.	The student uses processing skills and strategies with considerable effectiveness.	The student uses processing skills and strategies with a high degree of effectiveness.
Use of critical/creative thinking processes, skills, and strategies (e.g., <i>analysing, interpreting, problem solving, evaluating, forming and justifying conclusions on the basis of evidence, developing solutions, considering diverse perspectives</i>)	The student uses critical/creative thinking processes, skills, and strategies with limited effectiveness.	The student uses critical/creative thinking processes, skills, and strategies with some effectiveness.	The student uses critical/creative thinking processes, skills, and strategies with considerable effectiveness.	The student uses critical/creative thinking processes, skills, and strategies with a high degree of effectiveness.

Communication – The conveying of meaning through various forms

Categories	Level 1	Level 2	Level 3	Level 4
Expression and organization of ideas and information in oral, visual, and/or written forms (e.g., <i>diagrams, models, articles, project journals, reports</i>)	The student expresses and organizes ideas and information with limited effectiveness.	The student expresses and organizes ideas and information with some effectiveness.	The student expresses and organizes ideas and information with considerable effectiveness.	The student expresses and organizes ideas and information with a high degree of effectiveness.
Communication for different audiences (e.g., <i>peers, adults, community members</i>) and purposes (e.g., <i>to inform, to persuade</i>) in oral, visual, and/or written forms	The student communicates for different audiences and purposes with limited effectiveness.	The student communicates for different audiences and purposes with some effectiveness.	The student communicates for different audiences and purposes with considerable effectiveness.	The student communicates for different audiences and purposes with a high degree of effectiveness.

Use of conventions, vocabulary, and terminology of the discipline in oral, visual, and/or written forms (e.g., symbols, formulae, International System of Units)	The student uses conventions, vocabulary, and terminology of the discipline with limited effectiveness.	The student uses conventions, vocabulary, and terminology of the discipline with some effectiveness.	The student uses conventions, vocabulary, and terminology of the discipline with considerable effectiveness.	The student uses conventions, vocabulary, and terminology of the discipline with a high degree of effectiveness.
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Application – The use of knowledge and skills to make connections within and between various contexts

Categories	Level 1	Level 2	Level 3	Level 4
Application of knowledge and skills (e.g., concepts and processes; procedures related to the safe use of tools, equipment, materials, and technology; investigation skills) in familiar contexts	The student applies knowledge and skills in familiar contexts with limited effectiveness.	The student applies knowledge and skills in familiar contexts with some effectiveness.	The student applies knowledge and skills in familiar contexts with considerable effectiveness.	The student applies knowledge and skills in familiar contexts with a high degree of effectiveness.
Transfer of knowledge and skills (e.g., concepts and processes, safe use of equipment and technology, investigation skills) to new contexts	The student transfers knowledge and skills to new contexts with limited effectiveness.	The student transfers knowledge and skills to new contexts with some effectiveness.	The student transfers knowledge and skills to new contexts with considerable effectiveness.	The student transfers knowledge and skills to new contexts with a high degree of effectiveness.
Making connections within and between various contexts (e.g., connections between sciences; connections to everyday and real-life situations; connections among concepts within science and technology; connections involving use of prior knowledge and experience; connections among science and technology and other disciplines, including other STEM [science, technology, engineering, and mathematics] subjects)	The student makes connections within and between various contexts with limited effectiveness.	The student makes connections within and between various contexts with some effectiveness.	The student makes connections within and between various contexts with considerable effectiveness.	The student makes connections within and between various contexts with a high degree of effectiveness.

Proposing courses of practical action to deal with problems relating to our changing world	The student proposes courses of practical action of limited effectiveness.	The student proposes courses of practical action of some effectiveness.	The student proposes courses of practical action of considerable effectiveness.	The student proposes highly effective courses of practical action.
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The Achievement Chart for English, Grades 11 and 12

Knowledge and Understanding – Subject-specific content acquired in each course (knowledge), and the comprehension of its meaning and significance (understanding)

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
Knowledge of content (<i>e.g., forms of text; strategies used when listening and speaking, reading, writing, and viewing and representing; elements of style; literary terminology, concepts, and theories; language conventions</i>)	The student demonstrates limited knowledge of content.	The student demonstrates some knowledge of content.	The student demonstrates considerable knowledge of content.	The student demonstrates thorough knowledge of content.
Understanding of content (<i>e.g., concepts; ideas; opinions; relationships among facts, ideas, concepts, themes</i>)	The student demonstrates limited understanding of content.	The student demonstrates some understanding of content.	The student demonstrates considerable understanding of content.	The student demonstrates thorough understanding of content.

Thinking – The use of critical and creative thinking skills and/or processes

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
Use of planning skills (<i>e.g., generating ideas, gathering information, focusing research, organizing information</i>)	The student uses planning skills with limited effectiveness.	The student uses planning skills with some effectiveness.	The student uses planning skills with considerable effectiveness.	The student uses planning skills with a high degree of effectiveness.
Use of processing skills (<i>e.g., drawing inferences, interpreting, analysing, synthesizing, evaluating</i>)	The student uses processing skills with limited effectiveness.	The student uses processing skills with some effectiveness.	The student uses processing skills with considerable effectiveness.	The student uses processing skills with a high degree of effectiveness.

Use of critical/creative thinking processes (<i>e.g., oral discourse, research, critical analysis, critical literacy, metacognition, creative process</i>)	The student uses critical/creative thinking processes with limited effectiveness.	The student uses critical/creative thinking processes with some effectiveness.	The student uses critical/creative thinking processes with considerable effectiveness.	The student uses critical/creative thinking processes with a high degree of effectiveness.
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Communication – The conveying of meaning through various forms

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
Expression and organization of ideas and information (<i>e.g., clear expression, logical organization</i>) in oral, graphic, and written forms, including media forms	The student expresses and organizes ideas and information with limited effectiveness.	The student expresses and organizes ideas and information with some effectiveness.	The student expresses and organizes ideas and information with considerable effectiveness.	The student expresses and organizes ideas and information with a high degree of effectiveness.
Communication for different audiences and purposes (<i>e.g., use of appropriate style, voice, point of view</i>) in oral, graphic, and written forms, including media forms	The student communicates for different audiences and purposes with limited effectiveness.	The student communicates for different audiences and purposes with some effectiveness.	The student communicates for different audiences and purposes with considerable effectiveness.	The student communicates for different audiences and purposes with a high degree of effectiveness.
Use of conventions (<i>e.g., grammar, spelling, punctuation, usage</i>), vocabulary, and terminology of the discipline in oral, graphic, and written forms, including media forms	The student uses conventions, vocabulary, and terminology of the discipline with limited effectiveness.	The student uses conventions, vocabulary, and terminology of the discipline with some effectiveness.	The student uses conventions, vocabulary, and terminology of the discipline with considerable effectiveness.	The student uses conventions, vocabulary, and terminology of the discipline with a high degree of effectiveness.

Application – The use of knowledge and skills to make connections within and between various contexts

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
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Application of knowledge and skills <i>(e.g., literacy strategies and processes; literary terminology, concepts, and theories) in familiar contexts</i>	The student applies knowledge and skills in familiar contexts with limited effectiveness.	The student applies knowledge and skills in familiar contexts with some effectiveness.	The student applies knowledge and skills in familiar contexts with considerable effectiveness.	The student applies knowledge and skills in familiar contexts with a high degree of effectiveness.
Transfer of knowledge and skills <i>(e.g., literacy strategies and processes; literary terminology, concepts, and theories) to new contexts</i>	The student transfers knowledge and skills to new contexts with limited effectiveness.	The student transfers knowledge and skills to new contexts with some effectiveness.	The student transfers knowledge and skills to new contexts with considerable effectiveness.	The student transfers knowledge and skills to new contexts with a high degree of effectiveness.
Making connections within and between various contexts <i>(e.g., between the text and personal knowledge and experience, other texts, and the world outside school)</i>	The student makes connections within and between various contexts with limited effectiveness.	The student makes connections within and between various contexts with some effectiveness.	The student makes connections within and between various contexts with considerable effectiveness.	The student makes connections within and between various contexts with a high degree of effectiveness.

TAS2O - Technology and the Skilled Trades, Grade 10

Open

Issued: 2024

This hands-on course enables students to apply the engineering design process and other technological knowledge and skills introduced in earlier grades. Students will design and safely create prototypes, products, and/or services, working with tools and resources from various industries. As students develop their projects to address real-life problems, they will apply technological concepts such as quality control, and health and safety standards. Students explore opportunities for job skills programs and education and training pathways, including skilled trades, that can lead to a variety of careers.

Broad-based technology focus courses

School boards may offer the following broad-based technology focus courses related to Grade 10 Technology and the Skilled Trades:

- TGJ2O – Communications Technology and the Skilled Trades
- TEJ2O – Computer Technology and the Skilled Trades
- TCJ2O – Construction Technology and the Skilled Trades
- THJ2O – Green Industries and the Skilled Trades
- TXJ2O – Hairstyling and Aesthetics and the Skilled Trades
- TPJ2O – Health Care Technology and the Skilled Trades
- TFJ2O – Hospitality and Tourism and the Skilled Trades
- TMJ2O – Manufacturing Technology and the Skilled Trades
- TDJ2O – Technological Design and the Skilled Trades
- TTJ2O – Transportation Technology and the Skilled Trades

Prerequisite: None

Introduction

Preface

This is the curriculum policy for technological education, Grades 9 and 10, 2024. It supersedes *The Ontario Curriculum, Grades 9 and 10: Technological Education, 2009*. Effective beginning in the 2024–25 school year, all technological education courses for Grades 9 and 10 will be based on the expectations outlined in this curriculum policy.

In addition to the considerations outlined in this curriculum context, all of the general [“Program Planning”](#) sections apply to this curriculum. Educators should review and implement these general sections, as well as the components that appear below.

Vision and Goals of Technological Education, Grades 9 and 10

The vision of the technological education program in Grades 9 and 10 is for all students to acquire and develop the skills and knowledge related to technological education that will support them in contributing to the global economic, scientific, and societal innovations of tomorrow. This learning will enhance their ability to achieve success in secondary school, the workplace, postsecondary education or training, and daily life. A central component of this curriculum is safe, practical, hands-on, experiential learning that will support students in becoming technologically literate citizens. That is, they are able to understand, work with, and benefit from a range of technologies – skills that are necessary given the power, reach, and rapid evolution of technology. To succeed in today’s rapidly changing and technologically dependent society, students need to be effective problem solvers and critical thinkers, able to understand, question, anticipate, and respond to the implications of technological innovation. Students who pursue careers in technology will also need these high-level skills to develop solutions to technological challenges or to provide the services required in their chosen fields.

The goals of the technological education curriculum for Grades 9 and 10 are to enable students to:

- gain an understanding of the fundamental technological concepts underlying technological education through hands-on, project-based learning while developing technical skills;
- develop creative and flexible approaches to problem solving that will help them address challenges in various areas in everyday life;
- develop a level of technological proficiency that will allow them to be critical consumers and producers of technological solutions;
- explore the impact and development of technology, including emerging technological innovations;
- see themselves as capable and successful STEM (science, technology, engineering, and mathematics) learners and practitioners;
- explore career opportunities in technological fields and the skilled trades, and make connections that will help them take advantage of job skills programs and potential postsecondary pathways, including apprenticeship.

The Importance of Technological Education in the Curriculum

Technological innovation influences all areas of life, from the daily lives of individuals, to the work of business and government, to interactions on a global scale. It helps meet basic human needs and provides tools for improving people’s lives and exploring new challenges.

Technological education promotes the integration of learning across subject disciplines. For example, when students design a product, they explore the social or human need that the product addresses (social science), the scientific principles involved in its design and construction (science), its dimensions and shape (mathematics), and the aesthetic qualities of its design (the arts). When students consider which materials to use to create their product, they must consider sustainability and material costs. When they assess the impact that technological innovations have had – or may have – on society and the environment, students are exploring historical or current events and scientific or geographic concepts. When they consider how various technologies affect physical and mental health and well-being, they are looking into aspects of health and physical education. They apply literacy skills to communicate design ideas, produce reports summarizing technological projects, and write instructions for the use of the products they create.

Technological education also helps students develop research skills and fosters creativity, critical thinking, and problem solving. In addition, in its emphasis on innovation to meet human needs, it encourages global citizenship and promotes social, economic, and environmental responsibility.

The Importance of Technological Education in STEM Education

STEM education is the cross-curricular study of science, technology, engineering, and mathematics (STEM), and the application of those subjects in real-world contexts. As students engage in STEM education, they develop the [transferable skills](#) that they need to meet the demands of today's global economy and society.

Skills developed through STEM education include those related to problem solving, innovation, and engineering design. These skills are in high demand in today's changing world, as technology continues to impact all areas of society. In these courses, students use an engineering design process and associated skills to design, create, and test products or services. (For more on this, see the subsection "Engineering Design Process" under "The Program in Technological Education".) They also connect these skills to broad-based technology areas¹⁵ (see "Overview" in the same section) and to various careers,

¹⁵ The technological education curriculum in Grades 9–12 encompasses ten broad-based technology (BBT) areas, as follows:

- Communications Technology
- Computer Technology
- Construction Technology
- Green Industries
- Hairstyling and Aesthetics
- Health Care
- Hospitality and Tourism
- Manufacturing Technology
- Technological Design
- Transportation Technology

including the skilled trades, recognizing that skills developed in technological education have close connections to many areas of STEM.

Classroom activities focused on exploring solutions to real-world problems and on understanding practical applications of concepts can combine components from two or more STEM-related subjects and can include contexts related to the student's home and community or to various occupations, including the skilled trades. The integration of a number of STEM-related subjects can reinforce students' understanding of each subject and of the interrelationships among them. Similarly, reflecting on diverse perspectives engages students in a variety of creative and [critical thinking](#) processes that are essential for developing innovative, ethical, and effective responses to various needs.

The technological education curriculum for Grades 9 and 10 is designed to build on the foundation of knowledge and skills provided by the elementary science and technology curriculum. In this continuum, there is a similar emphasis on foundational knowledge and skills (fundamentals), engineering design, technological problem-solving skills and processes, and the relationship between technology, the environment, and society.

Learning by Doing: A Hands-On Approach to Technological Education

The core principle that underlies technological education is that *students learn best by doing*. This curriculum focuses on experiential, hands-on, project-driven approaches that involve students in problem solving as they develop knowledge and skills and gain experience in a variety of broad-based technology areas.

Experiential learning that takes place in the community, including job shadowing, job twinning, work experience, and cooperative education, provides students with the opportunity to enhance their learning. These experiences also support students by helping them make successful transitions to an initial postsecondary destination, including apprenticeship. There can also be significant benefits to communities in providing experiential learning opportunities for students. Community partners can take pride in contributing to the education of children and youth and in helping to develop Ontario's future workforce.

Teachers are encouraged to provide valuable experiential learning opportunities by connecting students with role models with diverse experiences. This may include presentations given by guest speakers, particularly from populations that are underrepresented in technological education and the skilled trades.

High schools across Ontario offer job skills programs to help young people recognize career paths in technological fields and the skilled trades. Examples of these job skills programs include Specialist High Skills Majors (SHSM), dual credit programs, the Ontario Youth Apprenticeship Program (OYAP) and Focused Apprenticeship Skills Training (OYAP-FAST), and cooperative education courses. For more information, see the "Program Planning" [section on experiential learning](#).

Fundamental Technological Concepts

This curriculum identifies fundamental technological concepts that inform design and production in various areas of technology. To address technological challenges and solve problems effectively, students need to take the full range of these concepts into account. As they progress through the technological education courses, students will come to understand these concepts more deeply, and to work with them creatively as they confront new challenges.

Definitions of Fundamental Technological Concepts

Aesthetics	The aspects of a product, process, or service that make it pleasing to the human senses.
Control	The means by which a device or process is activated or regulated. This includes automation, which involves using technology to make a device or process run on its own.
Environmental sustainability	The creation of products or services and use of resources in a way that allows present needs to be met without compromising the ability of future generations to meet their needs. An important related concept is that of <i>environmental stewardship</i> – the responsibility for the sustainable use and treatment of natural resources.
Ergonomics	The design of a product, process, or service in a way that takes the user's well-being with respect to its use or delivery into account – that is, in a way that minimizes discomfort, risk of injury, and expenditure of energy.
Creation	The act or process of assembling components and/or materials and resources to fabricate/build/create a product or service.
Function	The use for which a product, process, or service is developed.
Innovation	Original and creative thinking resulting in the effective design of a product or service.
Material	Any substance or item used in the creation of a product or delivery of a service.
Mechanism	A system of connected parts that allows a product to work or function.
Power and energy	The resource that enables a mechanism to perform work.
Safety	The care and consideration required to ensure that the product, process, or service will not cause harm.
Structure	The essential physical or conceptual parts of a product, process, or service, including the way in which the parts are constructed or organized.
Systems	The combinations of interrelated parts that make up a whole and that may be connected with other systems.

Project Management

While building technological concepts and skills, students will develop project management skills that will equip them to set goals, establish performance or evaluation criteria, assess risks, select and manage resources, organize and assign tasks, distribute responsibilities, manage deadlines and progress,

and lead others through the design process. Developing these skills supports their success in many broad-based technology areas and disciplines. Project management skills will support students in succeeding in any career path and help them thrive in a rapidly evolving and fast-paced world to become the proficient leaders of tomorrow.

The Program in Technological Education

Overview

The technological education curriculum in Grades 9–12 encompasses ten broad-based technology areas, as follows:

- Communications Technology
- Computer Technology
- Construction Technology
- Green Industries
- Hairstyling and Aesthetics
- Health Care
- Hospitality and Tourism
- Manufacturing Technology
- Technological Design
- Transportation Technology

To support educators in planning for learning in the context of the various broad-based technology areas, a separate [resource table](#) has been created that provides a sampling of industry-specific concepts and topics, skills and applications, and careers.

Starting with students entering Grade 9 in the 2024–25 school year, students are required to earn one compulsory credit in Grade 9 or 10 technological education (TAS1O, TAS2O, or a focus course) towards their Ontario Secondary School Diploma.

Any Grade 9 and 10 technological education course may be delivered as a half-credit course (see below), but may not be planned as a multiple-credit course.

Courses in Technological Education, Grades 9 to 12

The course information that appears below is in effect starting in the 2024–25 school year. *The Ontario Curriculum, Grades 11 and 12: Technological Education, 2009* remains in effect for those grades. All references to Grades 9 and 10 in *The Ontario Curriculum, Grades 11 and 12: Technological Education, 2009* have been superseded by the 2024 curriculum.

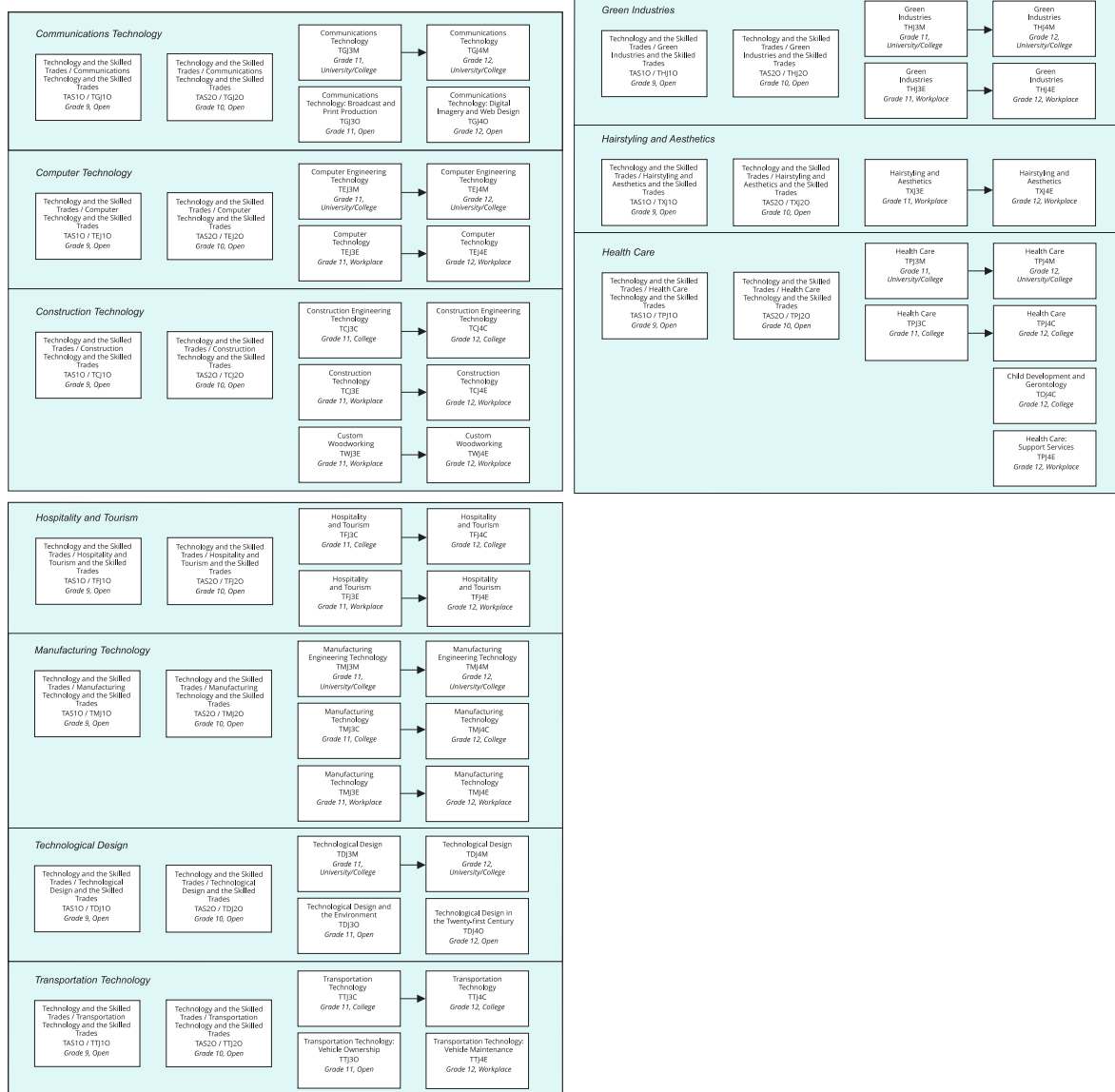
Grades 9 and 10

Grade	Course Name	Course Type	Course Code	Prerequisite
9	Technology and the Skilled Trades	Open	TAS10	None
10	Technology and the Skilled Trades	Open	TAS20	None

Note: Each of the courses listed above is worth one credit.

The Grade 9 and 10 courses may be adapted to create an additional course or courses that focus on any one of the broad-based technology areas listed above. For more information, see the section on focus courses below.

Prerequisite Charts for Technological Education, Grades 9–12



Technology and the Skilled Trades Courses (TAS10 and TAS20)

The Grade 9 and 10 Technology and the Skilled Trades courses (TAS10 and TAS20) build on the elementary science and technology and mathematics curricula and complement the Grade 9 and 10 science and mathematics courses. They are intended to introduce students to technological education in general, exposing them to a range of broad-based technology areas. Students in these courses work on projects that can encompass several different technological areas, using a range of equipment and resources suited to the various broad-based technology areas and learning environments.

Focus Courses

Schools may offer one or more focus courses in Grade 9 and/or 10 to explore groups of related occupations and industry sectors, including [industry-specific concepts and skills](#), within a particular broad-based technology area. For example, focus courses in construction technology enable students to acquire knowledge and skills related to carpentry, electrical/network cabling, heating and cooling, masonry, and plumbing.

School boards may offer the following broad-based technology focus courses related to Grade 9 Technology and the Skilled Trades:

- TGJ10 – Communications Technology and the Skilled Trades
- TEJ10 – Computer Technology and the Skilled Trades
- TCJ10 – Construction Technology and the Skilled Trades
- THJ10 – Green Industries and the Skilled Trades
- TXJ10 – Hairstyling and Aesthetics and the Skilled Trades
- TPJ10 – Health Care Technology and the Skilled Trades
- TFJ10 – Hospitality and Tourism and the Skilled Trades
- TMJ10 – Manufacturing Technology and the Skilled Trades
- TDJ10 – Technological Design and the Skilled Trades
- TTJ10 – Transportation Technology and the Skilled Trades

School boards may offer the following broad-based technology focus courses related to Grade 10 Technology and the Skilled Trades:

- TGJ20 – Communications Technology and the Skilled Trades
- TEJ20 – Computer Technology and the Skilled Trades
- TCJ20 – Construction Technology and the Skilled Trades
- THJ20 – Green Industries and the Skilled Trades
- TXJ20 – Hairstyling and Aesthetics and the Skilled Trades
- TPJ20 – Health Care Technology and the Skilled Trades
- TFJ20 – Hospitality and Tourism and the Skilled Trades
- TMJ20 – Manufacturing Technology and the Skilled Trades
- TDJ20 – Technological Design and the Skilled Trades
- TTJ20 – Transportation Technology and the Skilled Trades

The additional course(s) must be developed on the basis of the expectations outlined in this curriculum for Technology and the Skilled Trades, Grade 9 (TAS1O) and/or Technology and the Skilled Trades, Grade 10 (TAS2O), and must be assigned a specified name and course code as described above. For example, a Grade 9 course could be developed to focus on communications technology (TGJ10), or a Grade 10 course could be developed to focus on construction technology (TCJ20).

Regardless of the particular broad-based technology area on which a course is focused, students must be given the opportunity to achieve all of the expectations of the course outlined in this curriculum.

Students may take, and earn credit for, more than one technological education course in Grades 9 and 10. For example, a student could take two courses in Grade 10: Technology and the Skilled Trades (TAS2O) and Construction Technology and the Skilled Trades (TCJ2O), earning one credit for each.

Half-Credit Courses

The courses outlined in this curriculum are designed to be offered as full-credit courses. However, they may also be delivered as half-credit courses. Half-credit courses, which require a minimum of fifty-five hours of scheduled instructional time, must adhere to the following conditions:

- The two half-credit courses created from a full course must together contain all of the expectations of the full course.
- Students must successfully complete both parts of the course if it is to be used as a prerequisite for another course.
- The title of each half-credit course must include the designation Part 1 or Part 2. A half credit (0.5) will be recorded in the credit-value column of both the report card and the Ontario Student Transcript.

Boards will report all half-credit courses to the ministry annually in the School October Report.

Curriculum Expectations

The expectations identified for each course describe the skills and knowledge that students are expected to acquire, demonstrate, and apply in their projects and in various other activities on which their achievement is assessed and evaluated.

Mandatory learning is described in the overall and specific expectations of the curriculum.

Two sets of expectations – overall expectations and specific expectations – are listed for each *strand*, or broad area of the curriculum. In Grades 9 and 10, there are two strands: Strand A: Design Processes and Related Skills and Strand B: Technological Development, Impacts, and Careers. *Taken together, the overall and specific expectations represent the mandated curriculum.*

The *overall expectations* describe in general terms the skills and knowledge that students are expected to demonstrate by the end of the course. The *specific expectations* describe the expected skills and knowledge in greater detail. The specific expectations are organized under numbered subheadings, each of which indicates the strand and the overall expectation to which the group of specific expectations corresponds (e.g., “B2” indicates that the group relates to overall expectation 2 in Strand B). This organization is not meant to imply that the expectations in any one group are achieved independently of the expectations in the other groups, nor is it intended to imply that the learning associated with the expectations happens in a linear, sequential way. The numbered headings are used merely as an organizational structure to help teachers focus on particular aspects of knowledge, concepts, and skills as they develop various lessons and learning activities for students.

Teacher Supports

Specific expectations are accompanied by supports such as examples, teacher prompts, and/or instructional tips.¹⁶ Examples are meant to clarify the requirement specified in the expectation, illustrating the kind of skill or knowledge, the specific area of learning, the depth of learning, and/or the level of complexity that the expectation entails. Teacher prompts are sample guiding questions and considerations that can lead to discussions and promote deeper understanding. Instructional tips suggest instructional strategies and authentic contexts for the effective modelling, practice, and application of technological education concepts.

Teacher supports, such as examples, teacher prompts, and instructional tips, are optional supports that educators can draw on to support teaching and learning, in addition to developing their own supports that reflect a similar level of complexity. Facilitating learning towards the achievement of the specific expectations must always be responsive to the diversity of the learners in the classroom. Universal Design for Learning that removes barriers for students, differentiated instruction, and flexible strategies are intended to support learners towards the successful demonstration of the expectations.

Strands in the Technological Education Curriculum for Grades 9 and 10

The overall and specific expectations for each course in the technological education curriculum for Grades 9 and 10 are organized in two distinct but related strands. As students move up through the grades, the expectations within these strands will increase in complexity and depth. These strands are as follows:

- **Strand A: Design Processes and Related Skills** focuses on hands-on and project-based learning to build foundational skills, including: the acquisition and application of fundamental technological skills; various design processes, including the engineering design process; the creation of prototypes for products or services; and health and safety practices.
- **Strand B: Technological Development, Impacts, and Careers** focuses on foundational learning related to the nature of technology and innovation, including: transferable skills; cross-curricular STEM connections; the impact of technology; contributions to technological innovations made by members of diverse groups and communities in Canada; and opportunities and careers, including skilled trades.

¹⁶ The teacher supports will be made available at a later date, after the issuing of the curriculum expectations and the curriculum context.

Engineering Design Process

In many technological fields, open-ended problem-solving processes that involve the full cycle of planning and developing products or services to meet identified needs are often referred to as “design processes”. A design process is an iterative process that may involve revisiting a prior stage as new information is acquired about the problem being investigated, or as a better understanding is developed of the person or people for whom a solution is being designed. The entire process may even be restarted, or repeated, when one approach proves unsuccessful.

Since students will be seeking solutions to problems that will impact others, ethical considerations as well as the perspectives and needs of a variety of individuals and communities should be considered throughout the process. Students can conduct interviews with end-users, or they can research individuals or communities that may be affected by potential solutions. Their approach should be empathetic, and students should consider various perspectives, as well as factors such as environmental sustainability and usability, throughout the process.

The engineering design process described below involves students initiating and planning solutions, and “performing” or carrying out their plan by designing and creating prototypes of their solutions. A prototype is a functional model of a product or service that can be used to demonstrate the solution and obtain feedback from the prospective end-user. This information is used to test and validate design choices that students have made. Students analyze and refine their prototypes based on the information they have collected. Communication and documentation occur throughout the process, using appropriate vocabulary and forms for a variety of purposes. The end product of the engineering design process might not be a tangible object; it might instead be a technological process or service.

There is no single design process, but rather a range of practices and design processes that are followed when designing solutions or developing projects. The engineering design process in technological education is similar to the process described in elementary science and technology and Grade 9 science, with a particular focus on students’ thinking critically about *how* they will be creating their prototype. At the secondary level, technological education students should be considering the methods, approaches, and techniques they will use in creating their prototype, as well as time constraints, environmental impact, cost of materials and resources, and tools and materials required. These considerations inform the development of the process or method by which they will create their prototype.

Communication and documentation are areas of focus throughout the engineering design process in technological education. Actively engaging in technical documentation and communication allows students to record, reflect, and communicate throughout the development process of their projects. These skills extend beyond secondary technological education as students explore careers and postsecondary pathways in technological fields, including the skilled trades.

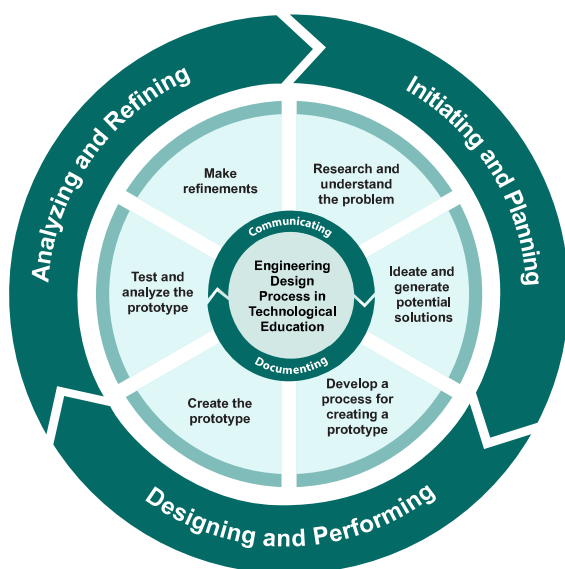
Students and teachers may find the need to emphasize specific aspects of the engineering design process provided, or to make substitutions with components of processes that they may find elsewhere. Students and teachers may even find other engineering design processes that they may want to work with, and a comparison of various processes may prove beneficial for students and teachers.

Appropriate entry points into the engineering design process and the specific components of the process that are focused on may depend on student readiness. Prior experience and knowledge, as well as access to resources, the context of the learning, and the amount of time available, may also be factors; therefore, educators may need to provide multiple entry points to engage all students in the learning.

The engineering design process provided here allows students to engage with important technological concepts and skills within curriculum expectations as they develop the transferable skills and cross-curricular concepts that embody STEM education.

The following diagram summarizes the engineering design process in technological education and shows how its components relate to the skills of *initiating and planning*; *designing and performing*; *analyzing and refining*; and *communicating and documenting*. The components are described in more detail below.

Engineering Design Process in Technological Education



Initiating and Planning

Research and understand the problem

- Analyze the context and clearly define the problem or challenge.
- Identify the users affected by the problem.
- Listen closely to those affected by the problem and use empathy to understand their experiences, perspectives, and concerns.
- Conduct research to determine design considerations, financial or other constraints, and availability or suitability of materials.
- Review related problems and solutions to these problems.
- Identify issues related to sustainability and to health and safety.

Ideate and generate potential solutions

- Brainstorm several ideas and potential solutions.
- Review potential solutions, considering related research, problems, and solutions.
- Develop specific evaluation criteria and constraints, and evaluate potential solutions based on these criteria and constraints.
- Consider the end-users and those impacted by potential solutions, taking into consideration their experiences, perspectives, and concerns.
- Consider applying related and existing solutions (or some aspects of them) to the identified problem.
- Consider developing new solutions that are different from existing solutions.
- Refine or combine potential solutions.
- Select the most appropriate solution, based on established criteria.

Designing and Performing

Develop a process for creating a prototype

- Plan the design of the solution, considering the required stages and available materials and other resources.
- Consider the economic, environmental, ethical, and health and safety concerns related to the potential design.
- Consider the key components of the design, and ensure that they can be effectively produced.

Create the prototype

- Select the tools and equipment required to create the prototype safely.
- Carry out their plan to create the model or prototype.
- Review the steps and criteria from the plan and modify as needed.
- Document the modifications made.

Analyzing and Refining

Test and analyze the prototype

- Develop tests to evaluate the solution using developed evaluation criteria.
- Conduct tests in a variety of contexts, including in controlled and in real-world environments and with various potential end-users.
- Obtain feedback on the prototype from potential end-user(s) and others, including teachers, classmates, friends, family members, and/or community members.
- Record observations and data.

Make refinements

- Analyze results from testing to determine what changes should be made to the prototype to enhance the end-user experience.
- Considering the results of testing, review initial resources, existing knowledge, and other brainstormed ideas to improve upon the design.
- Consider additional components, materials, equipment, or time needed.
- Taking into account all information, refine the prototype and develop a finished product or service.

Communicating and Documenting¹⁷

- Document the development process.
- Describe challenges encountered in the development process and possible ways to address these challenges.
- Choose a form or medium for communication that is appropriate for the intended audience.
- Identify the important information and components of the solution or project to share, and develop a draft or plan for the presentation or demonstration, using appropriate vocabulary.
- Present or demonstrate the final prototype, finished product, or service to users and/or stakeholders.

Problem Solving in Technological Education

An approach to learning that emphasizes problem solving is the best way to prepare students for the challenges they will face in the world beyond school. Projects or tasks may not always be clearly defined or have prescribed solutions. Students who have a strong background in problem solving will be more confident and better equipped to address new challenges in a variety of contexts.

Learning through problem solving will help students appreciate that all challenges – whether large or small, complex or simple – are most effectively resolved when approached systematically, using a simple method or a more comprehensive process, depending on the nature of the problem.

The range of challenges students encounter in technological education is wide and varied. At one end are simple problems for which there is likely to be only one solution – for example, substituting a part to fix an obvious fault. At the other end are complex challenges – for example, developing a solution to an identified human need – for which there could be various solutions and which call for a detailed process that may involve consultations with stakeholders to clearly define the problem and determine criteria for its solution, and the design and testing of several potential solutions. In many cases, the nature of

¹⁷ Communication and documentation are done throughout the engineering design process.

the problem, and the problem-solving process required to solve it, fall somewhere between these two extremes.

Teachers can guide students through problem solving by helping them understand the nature and scope of a problem and the type of approach or method best suited to address it. They can also remind students that there is often more than one solution, give them the freedom to explore ideas, and encourage them to retrace steps and persist in their efforts when they encounter obstacles.

Problem-Solving Methods and Approaches

Problem-solving processes share at least some of a number of systematic steps – for example, identifying the problem, analyzing the situation, considering possible solutions, selecting the best solution, testing and evaluating the effectiveness of the solution, and reviewing or repeating steps as necessary to improve the solution. Among the various problem-solving methods and approaches that may be employed to address the range of problems students will encounter in technological education are those listed below. This list is not comprehensive, and may be supplemented by various other methods in the learning environment.

Diagnostics

An example of a diagnostic problem-solving method is troubleshooting an engine fault in an automobile. After identifying the general problem, the technician would run tests to pinpoint the fault. The test results would be used either as a guide for further testing or for replacement of a part, which would also need to be tested. This process continues until the solution is found and the car is running properly.

Reverse Engineering

“Reverse engineering” is the process of carefully tracing the workings of a product or process and identifying its fundamental component.

Extreme Cases

Considering “extreme cases” – envisioning the problem in a greatly exaggerated or greatly simplified form, or testing using an extreme condition – can often help to pinpoint a problem. An example of the extreme-case method is purposely inputting an extremely high number to test a computer program.

Divide and Conquer

“Divide and conquer” is the technique of breaking down a problem into subproblems, then breaking the subproblems down even further until each of them is simple enough to be solved. Divide and conquer may be applied to allow groups of students to tackle subproblems of a larger problem, or when a problem is so large that its solution cannot be visualized without breaking it down into smaller components.

Parts Substitution

Perhaps the most basic of all the problem-solving methods, “parts substitution” simply requires that parts be substituted until the problem is solved. Although it is not the most efficient method of problem solving, there may be no other alternative if tests do not indicate what could be causing the problem.

Trial and Error

The trial-and-error method involves trying different approaches until a solution is found. It is often used as a last resort when other methods have been exhausted.

Program Planning and Cross-Curricular and Integrated Learning

Educators consider many factors when planning a technological education program that cultivates the best possible environment in which all students can maximize their learning. This section highlights important areas of focus that educators should consider, including areas of cross-curricular and integrated learning, as they plan effective, inclusive, and accessible technological education programs. In addition, all of the general [“Program Planning”](#) sections on the Curriculum and Resources website apply to this curriculum.

Instructional Approaches in Technological Education

Technological education involves knowing and doing, and teaching and learning strategies should address both areas, with an emphasis on doing. Teachers should use projects as a means of achieving course expectations, and students should be provided with a combination of information and hands-on experiences that will prepare them to make informed choices about the use of various technologies, to use technology safely and effectively, and to solve technological problems.

Students learn best when they are engaged in learning in a variety of ways. When students are engaged in active and experiential learning strategies, they tend to retain knowledge for longer periods and to develop, acquire, and integrate key skills more completely. Technological education courses lend themselves to a wide range of strategies in that they require students to discuss issues, conduct research, solve problems, plan solutions, develop and create products and/or services, think critically, and work cooperatively.

Programs in technological education are based on hands-on activities and should involve an open and collaborative approach to teaching that reflects students’ interests and aspirations. Activities should be designed to include both individual and team approaches, as technological projects in the workplace often require individuals to work collaboratively while undertaking a variety of roles, tasks, and responsibilities. Students should be given opportunities to work both independently and with teacher direction, and to learn through the study of examples followed by practice. There is no single correct way to teach or to learn, and the strategies used in the learning environment should vary according to the curriculum expectations and the needs of the students. Problem solving and/or an engineering design process are essential tools in broad-based technological education. Teachers should work collaboratively with colleagues to plan and deliver the technological education curriculum. Individual

teachers can contribute their expertise in particular areas of technology to ensure the successful implementation of the curriculum.

Some of the teaching and learning strategies that are suitable to material taught in technological education employ scaffolding. Scaffolding is an instructional approach that involves breaking down tasks so that students can concentrate on specific, manageable objectives and gradually build understanding and skill, with the aid of modelling by the teacher and ample opportunity for practice. Scaffolding provides students with a supportive structure within which to learn.

When students work collaboratively, they often accomplish more than when they work individually. Group activities, when used in a structured way, can enhance learning and foster positive attitudes. When working in a team, each student should have a specific role and be actively involved in the task. It is important to give students opportunities to take on different roles, from one project to another or in the course of a large project.

Students' attitudes towards technological education can have a significant effect on their achievement of expectations. Teaching methods and learning activities that encourage students to recognize the value and relevance of what they are learning for work and their lives beyond school will go a long way towards motivating students to work and learn effectively.

The study of current events related to technologies in various industries, including emerging technologies, should inform the technological education curriculum, enhancing both the relevance and the immediacy of the program. Discussion of current events related to various technologies and inclusion of these topics in daily lessons will stimulate students' interest and curiosity and also help them connect what they are learning in class with real-world events or situations. The study of events in industrial sectors and technological developments in the world needs to be thought of not as a separate topic removed from the program but as an effective instructional strategy for implementing many of the applicable expectations found in the curriculum.

Universal Design for Learning (UDL) and Differentiated Instruction (DI) in Technological Education

Students in every classroom, including in technological education, vary in their identities, experiences, personal interests, learning profiles, and readiness to learn new concepts and skills. Universal Design for Learning (UDL) and differentiated instruction (DI) are robust and powerful approaches to designing assessment and instruction to engage all students in technological education tasks that develop conceptual understanding. UDL and DI can be used in combination to help teachers respond effectively to the strengths and needs of all students.

The aim of the UDL framework is to assist teachers in designing technological education learning opportunities and environments that provide all students with inclusive, flexible, and equitable access to the curriculum. Teachers of technological education take into account students' diverse learner profiles to design lessons and tasks that offer individual choice, ensuring relevance and authenticity, providing

individual choice and graduated levels of challenge, and fostering collaboration in the learning environment. Teachers also represent technological concepts and information in multiple ways to help students become resourceful and knowledgeable learners. Technological education teachers can use a variety of media presentation or demonstration methods to ensure that within each lesson, students are provided with alternatives for auditory and visual information. For example, when students are learning how to use a tool in the classroom, teachers can share a video that outlines its proper use, conduct an in-person demonstration of its use, and allow students to demonstrate their understanding of its use.

Students have a diverse set of learning needs and a range of cultural experiences and identities. Teachers incorporate culturally relevant teaching, centring students' cultural backgrounds, into learning experiences to ensure that learning is authentic, meaningful, and relevant to students. For example, teachers could use UDL principles to make space for First Nations, Métis, and Inuit perspectives and contributions in the learning for *all* students, including Indigenous students. These planned opportunities facilitate the exploration of various meaningful and engaging forms of resources, including audio recordings, digital archives, virtual exhibits, or interviews with First Nations, Métis, and Inuit community members.

Designing technological education assignments and tasks through UDL from the start enables students to find their own entry point to the learning. Teachers use a tiered approach to ensure responsive, timely, and effective instruction that improves student learning while maintaining high expectations and rigour for all students. To support learners as they focus strategically on their learning goals, teachers create an environment in which learners can express themselves using a range of strengths. For example, teachers can vary ways in which students can respond and demonstrate their understanding of technological concepts, such as creating a video, doing a demonstration, or creating a set of instructions to illustrate the use of a technique. Teachers can also support students in developing and demonstrating project management skills related to their learning in technological education.

While UDL provides teachers with broad principles for planning instruction and learning experiences in technological education for a diverse group of students, differentiated instruction (DI) allows them to address specific skills and learning needs on a case-by-case basis. DI is student centred and involves a strategic blend of whole-class, small-group, and individual learning activities to suit students' differing strengths, interests, and levels of readiness to learn.

Attending to students' varied readiness for learning in technological education is an important aspect of differentiated instruction. Students need support in aiming higher, developing belief in excellence, and co-creating problem-based tasks of increasing complexity. To make some concepts more accessible, teachers can employ strategies such as offering students choice and providing open-ended problems that are based on situations relevant to their everyday lives.

Universal Design for Learning and differentiated instruction are integral aspects of an inclusive technological education program with the goal of achieving equity in technological education. More information on these approaches can be found in the ministry publication [*Learning for All: A Guide to Effective Assessment and Instruction for All Students, Kindergarten to Grade 12* \(2013\)](#).

Health and Safety in Technological Education

In Ontario, various laws, including the [Education Act](#), the [Occupational Health and Safety Act \(OHSA\)](#), [Ryan's Law \(Ensuring Asthma Friendly Schools\), 2015](#), and [Sabrina's Law, 2005](#), collectively ensure that school boards provide a safe and productive learning and work environment for both students and employees. Under the Education Act, teachers are required to ensure that all reasonable safety procedures are carried out in the programs and activities for which they are responsible. Teachers should always model safe practices; communicate safety requirements to students in accordance with school board policies, Ministry of Education policies, and any applicable laws; and encourage students to assume responsibility for their own safety and the safety of others.

Concern for safety must be an integral part of instructional planning and implementation. Teachers are encouraged to review:

- their responsibilities under the Education Act;
- their rights and responsibilities under the Occupational Health and Safety Act;
- their school board's health and safety policy for employees;
- their school board's policies and procedures relating to student health and safety (e.g., those related to concussions; prevalent medical conditions such as anaphylaxis, asthma, diabetes, and/or epilepsy; outdoor education excursions);
- relevant provincial subject association guidelines and standards for student health and safety;
- any additional mandatory requirements, particularly for higher-risk activities (e.g., field trips, workplaces), including requirements for approvals (e.g., from the supervisory officer), permissions (e.g., from parents¹⁸), and/or qualifications.

Wherever possible, potential risks should be identified and procedures developed to prevent or minimize, and respond to, incidents and injuries. School boards provide and maintain safe equipment, facilities, materials, and tools as well as providing qualified instruction. In safe learning environments, teachers will:

- be aware of up-to-date safety information;
- plan activities with safety as a primary consideration;
- inform students and parents of risks involved in activities;
- observe students to ensure that they are following safe practices, including the wearing of personal protective equipment;
- have a plan in case of emergency;
- show foresight by taking reasonable precautions to prevent health and safety accidents;
- act quickly.

¹⁸ The word *parent(s)* is used on this website to refer to parent(s) and guardian(s). It may also be taken to include caregivers or close family members who are responsible for raising the child.

To carry out their responsibilities with regard to safety, it is important not only that teachers have concern for their own safety and that of students, but also that they have:

- the knowledge necessary to safely use the materials and tools and follow the procedures involved in technological education;
- the skills needed to perform tasks safely.

Students should be made aware that health and safety is everyone's responsibility – at home; at school; in the community, including in the natural environment; and while visiting, and participating in experiential learning in, workplace settings. Teachers should ensure that students have the knowledge and skills needed for safe participation in all learning activities. Students must be able to demonstrate knowledge of the equipment, facilities, materials, and tools being used and the procedures necessary for their safe use.

Students demonstrate that they have the knowledge, skills, and safety mindset required for participation in technological education activities when they:

- maintain a well-organized and uncluttered workspace;
- follow established safety procedures;
- identify possible safety concerns and report them to the teacher;
- suggest and implement appropriate safety procedures;
- carefully follow the instructions and example of the teacher;
- focus on the task at hand, eliminating possible distractions;
- consistently show care and concern for their safety and that of others.

An important part of the technological education program is that students select appropriate equipment, materials, and tools for their projects. Schools and boards should collaborate to ensure that students have access to the necessary facilities, equipment, materials, tools, and appropriate safety training to support their learning and maintain a safe learning environment.

Learning outside the classroom, such as on field trips, can provide a meaningful and authentic dimension to students' learning experiences. Teachers must plan these activities carefully in accordance with their school board's relevant policies and procedures and in collaboration with other school board staff (e.g., the principal, board lead, supervisory officer) to ensure students' health and safety.

The information provided in this section is not exhaustive.

All aspects of the learning environment must comply with relevant school board, municipal, provincial, or federal health and safety legislation, including the following:

- the Ontario Occupational Health and Safety Act
- the Ontario Workplace Safety and Insurance Act
- the Workplace Hazardous Materials Information System (WHMIS 2015)
- the Food and Drugs Act
- the Ontario Health Protection and Promotion Act

- the Ontario Building Code
- local by-laws

Teachers should make use of all available and relevant resources to make students sufficiently aware of the importance of health and safety.

Innovations and Emerging Technology

A central focus of Strand B of these courses is learning related to innovations and emerging technologies, including the social, cultural, economic, environmental, and ethical issues related to their development and use. These can be engaging topics that capture the imagination of students as they consider exciting innovations in technology and imagine themselves playing a role in the development and application of these innovations, contributing to a hopeful and exciting future.

Students also analyze contributions to, and innovations in, technology by people from various communities, including First Nations, Métis, and Inuit communities in Canada. As students engage with this learning, they are empowered to consider that they can help shape the future in a positive way, potentially contributing to the development of future innovations by pursuing careers or further education in technological education or other STEM-related areas, including skilled trades.

Skilled Trades

A *skilled trade* is a career path that requires a particular skill set and specialty knowledge and training, and is usually hands-on. There are more [than 140 skilled trades in Ontario](#). Many skilled trades workers apply STEM-related concepts as they construct buildings; build and maintain infrastructure for transportation, communications, and utilities; or provide a range of professional services.

Throughout this curriculum, students will investigate how fundamental technological concepts and skills can be used in other disciplines and in real-world applications. Students also explore ways in which various industries are changing as a result of technological innovations. Career-related expectations provide opportunities for students to connect concepts and skills associated with this curriculum to potential postsecondary education and career pathways, including in the skilled trades.

Assessment and Evaluation of Student Achievement

[*Growing Success: Assessment, Evaluation, and Reporting in Ontario Schools, First Edition, Covering Grades 1 to 12, 2010*](#) sets out the Ministry of Education's assessment, evaluation, and reporting policy. The policy aims to maintain high standards, improve student learning, and benefit all

students, parents¹⁹, and teachers in elementary and secondary schools across the province. Successful implementation of this policy depends on the professional judgement of teachers at all levels as well as their high expectations of all students, and on their ability to work together and to build trust and confidence among parents and students.

Major aspects of assessment, evaluation, and reporting policy are summarized in the general “[Assessment and Evaluation](#)” section that applies to all curricula. The key tool for assessment and evaluation in technological education, Grades 9 and 10 – the achievement chart – is provided below.

The Achievement Chart for Technological Education, Grades 9 and 10

The achievement chart identifies four [categories of knowledge and skills](#) and four [levels of achievement](#) in technological education, Grades 9 and 10. (For important background, see “[Content Standards and Performance Standards](#)” in the general Assessment and Evaluation section.)

Knowledge and Understanding – Subject-specific content acquired in each course (knowledge), and the comprehension of its meaning and significance (understanding)

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
Knowledge of content (e.g., facts; use and function of equipment and tools; technical terminology; materials; concepts, processes, safety procedures)	The student demonstrates limited knowledge of content.	The student demonstrates some knowledge of content.	The student demonstrates considerable knowledge of content.	The student demonstrates thorough knowledge of content.
Understanding of content (e.g., safety procedures, fundamental technological concepts, processes, industry standards)	The student demonstrates limited understanding of content.	The student demonstrates some understanding of content.	The student demonstrates considerable understanding of content.	The student demonstrates thorough understanding of content.

Thinking – The use of critical and creative thinking skills and/or processes

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
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¹⁹ The word *parent(s)* is used on this website to refer to parent(s) and guardian(s). It may also be taken to include caregivers or close family members who are responsible for raising the child.

Use of planning skills (e.g., identifying a need or problem; generating and evaluating ideas; selecting strategies, tools, and resources; scheduling; budgeting)	The student uses planning skills with limited effectiveness.	The student uses planning skills with some effectiveness.	The student uses planning skills with considerable effectiveness.	The student uses planning skills with a high degree of effectiveness.
Use of processing skills (e.g., analyzing and interpreting information, forming conclusions)	The student uses processing skills with limited effectiveness.	The student uses processing skills with some effectiveness.	The student uses processing skills with considerable effectiveness.	The student uses processing skills with a high degree of effectiveness.
Use of critical/creative thinking processes (e.g., engineering design, service design, problem-solving, decision-making, diagnostic, and quality assurance processes)	The student uses critical/creative thinking processes with limited effectiveness.	The student uses critical/creative thinking processes with some effectiveness.	The student uses critical/creative thinking processes with considerable effectiveness.	The student uses critical/creative thinking processes with a high degree of effectiveness.

Communication – The conveying of meaning through various forms

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
Expression and organization of ideas and information (e.g., clarity, logic, coherence) in oral, non-verbal, visual, and/or written forms, including digital and media forms (e.g., demonstrations, technical descriptions/instructions, presentations, reports, flowcharts)	The student expresses and organizes ideas and information with limited effectiveness.	The student expresses and organizes ideas and information with some effectiveness.	The student expresses and organizes ideas and information with considerable effectiveness.	The student expresses and organizes ideas and information with a high degree of effectiveness.
Communication for different audiences (e.g., peers, clients, suppliers, colleagues, supervisors, the public) and purposes (e.g., to inform, to persuade, to collaborate) in oral, non-verbal, visual, and/or written forms, including digital and media forms	The student communicates for different audiences and purposes with limited effectiveness.	The student communicates for different audiences and purposes with some effectiveness.	The student communicates for different audiences and purposes with considerable effectiveness.	The student communicates for different audiences and purposes with a high degree of effectiveness.

Use of conventions (<i>e.g., standards, symbols, units of measurement, acronyms</i>), industry-related vocabulary, and terminology of the discipline in oral, non-verbal, visual, and/or written forms, including digital and media forms	The student uses conventions, vocabulary, and terminology with limited effectiveness.	The student uses conventions, vocabulary, and terminology with some effectiveness.	The student uses conventions, vocabulary, and terminology with considerable effectiveness.	The student uses conventions, vocabulary, and terminology with a high degree of effectiveness.
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Application – The use of knowledge and skills to make connections within and between various contexts

Categories	50–59% (Level 1)	60–69% (Level 2)	70–79% (Level 3)	80–100% (Level 4)
Application of knowledge and skills (<i>e.g., manipulation of materials; application of concepts and processes; safe use of tools, equipment, technology, and techniques</i>) in familiar contexts	The student applies knowledge and skills in familiar contexts with limited effectiveness.	The student applies knowledge and skills in familiar contexts with some effectiveness.	The student applies knowledge and skills in familiar contexts with considerable effectiveness.	The student applies knowledge and skills in familiar contexts with a high degree of effectiveness.
Transfer of knowledge and skills (<i>e.g., manipulation of materials; application of concepts and processes; safe use of tools, equipment, technology, and techniques</i>) to new contexts	The student transfers knowledge and skills to new contexts with limited effectiveness.	The student transfers knowledge and skills to new contexts with some effectiveness.	The student transfers knowledge and skills to new contexts with considerable effectiveness.	The student transfers knowledge and skills to new contexts with a high degree of effectiveness.
Making connections within and between various contexts (<i>e.g., connections to everyday personal situations; connections to social, economic, environmental, ethical, and cultural issues; connections between technological education and other disciplines; connections to potential careers and related postsecondary pathways, including apprenticeship</i>)	The student makes connections within and between various contexts with limited effectiveness.	The student makes connections within and between various contexts with some effectiveness.	The student makes connections within and between various contexts with considerable effectiveness.	The student makes connections within and between various contexts with a high degree of effectiveness.

Criteria and Descriptors for Technological Education, Grades 9 and 10

To guide teachers in their assessment and evaluation of student learning, the achievement chart provides “criteria” and “descriptors” within each of the four categories of knowledge and skills.

A set of criteria is identified for each category in the achievement chart. The criteria are subsets of the knowledge and skills that define the category. The criteria identify the aspects of student performance that are assessed and/or evaluated, and they serve as a guide to what teachers look for. In Technological Education, Grades 9 and 10, the criteria for each category are as follows:

Knowledge and Understanding

- knowledge of content (e.g., facts; use and function of equipment and tools; technical terminology; materials; concepts, processes, safety procedures)
- understanding of content (e.g., safety procedures, fundamental technological concepts, processes, industry standards)

Thinking

- use of planning skills (e.g., identifying a need or problem; generating and evaluating ideas; selecting strategies, tools, and resources; scheduling; budgeting)
- use of processing skills (e.g., analyzing and interpreting information, forming conclusions)
- use of critical/creative thinking processes (e.g., engineering design, service design, problem-solving, decision-making, diagnostic, and quality assurance processes)

Communication

- expression and organization of ideas and information (e.g., clarity, logic, coherence) in oral, non-verbal, visual, and/or written forms, including digital and media forms (e.g., demonstrations, technical descriptions/instructions, presentations, reports, flowcharts)
- communication for different audiences (e.g., peers, clients, suppliers, colleagues, supervisors, the public) and purposes (e.g., to inform, to persuade, to collaborate) in oral, non-verbal, visual, and/or written forms, including digital and media forms
- use of conventions (e.g., standards, symbols, units of measurement, acronyms), industry-related vocabulary, and terminology of the discipline in oral, non-verbal, visual, and/or written forms, including digital and media forms

Application

- application of knowledge and skills (e.g., manipulation of materials; application of concepts and processes; safe use of tools, equipment, technology, and techniques) in familiar contexts
- transfer of knowledge and skills (e.g., manipulation of materials; application of concepts and processes; safe use of tools, equipment, technology, and techniques) to new contexts
- making connections within and between various contexts (e.g., connections to everyday personal situations; connections to social, economic, environmental, ethical, and cultural issues;

connections between technological education and other disciplines; connections to potential careers and related postsecondary pathways, including apprenticeship)

Descriptors indicate the characteristics of the student's performance, with respect to a particular criterion, on which assessment or evaluation is focused. *Effectiveness* is the descriptor used for each criterion in the Thinking, Communication, and Application categories. What constitutes effectiveness in any given performance task will vary with the particular criterion being considered. Assessment of effectiveness may therefore focus on a quality such as appropriateness, clarity, accuracy, precision, logic, relevance, significance, fluency, flexibility, depth, or breadth, as appropriate for the particular criterion.

Expectations by strand

A. Design Processes and Related Skills

Students engage in an engineering design process throughout this strand.

Overall expectations

By the end of this course, students will:

A1. Initiating and Planning

demonstrate an understanding of fundamental technological concepts and related skills by initiating and planning projects

Specific expectations

By the end of this course, students will:

A1.1 apply an understanding of [fundamental technological concepts](#), and evaluate their significance in developing products and/or services in a variety of [broad-based technology areas](#)

A1.2 apply an understanding of fundamental technological concepts, design considerations, and science, technology, engineering, and mathematics (STEM) concepts as appropriate in developing projects involving the creation of products and/or services

A1.3 investigate design considerations, including accessibility requirements, that are relevant to developing projects, and identify those that are essential to various users or communities

A1.4 communicate design ideas for various purposes and audiences, using appropriate industry terminology and industry-standard formats and techniques

A1.5 establish and justify evaluation criteria for products and/or services being developed, including both qualitative and quantitative measures, making connections to relevant fundamental technological concepts

A1.6 investigate and describe project management skills and approaches that are relevant to developing products and/or services, and explain how they will use these skills in their own projects

A1.7 collect and synthesize information from a variety of sources, including people with diverse perspectives and from various communities, such as First Nations, Métis, and Inuit, to inform their projects

A2. Designing and Performing

develop projects that involve creating products and/or services, using a variety of resources and techniques, and record the development of their projects

Specific expectations

By the end of this course, students will:

A2.1 use project management skills to develop a process to create a product and/or service

A2.2 identify factors that could impact the development of their projects and apply appropriate strategies to increase the probability of a positive outcome

A2.3 select materials and other resources based on their properties or characteristics, including sustainability, and justify their use in the creation of products and/or services

A2.4 select, use, and maintain tools and equipment appropriately as part of creating products and/or delivering services

A2.5 use a variety of industry-related documents to guide the creation of products and/or the delivery of services as part of their projects

A2.6 create products and/or deliver services, documenting their development process using appropriate industry terminology

A2.7 select appropriate units of measure and tools to make accurate measurements using relevant measurement systems, such as the metric and imperial systems, and converting between systems and units

A3. Analyzing and Refining

evaluate and refine processes, products, and/or services

Specific expectations

By the end of this course, students will:

A3.1 identify challenges they encounter in the process of developing their projects and apply critical thinking skills to address these challenges and minimize the probability of their reoccurrence

A3.2 identify various industry-relevant performance standards and quality control methods

A3.3 analyze the performance of products and/or service delivery using quality control methods

A3.4 refine the design of products and/or services based on an analysis of data collected throughout the development process

A3.5 communicate project-related challenges, performance analyses, and refinements for a specific audience, using appropriate formats and terminology

A4. Applying Health and Safety Principles

apply an understanding of [health and safety](#) practices and procedures when using materials, tools, and equipment

Specific expectations

By the end of this course, students will:

A4.1 describe relevant health and safety regulations for a variety of settings, including mandated roles and responsibilities

A4.2 identify hazards in their environment, and apply strategies to minimize risks

A4.3 use tools and equipment safely, including using personal protective equipment and safety devices according to safety standards and regulations, as appropriate

A4.4 follow practices that support physical and mental health and well-being

A4.5 follow proper procedures for the safe handling, storage, and disposal of materials and waste products

A4.6 demonstrate a [safety mindset](#) by making safety a priority at all times and by engaging in industry-specific safety procedures

B. Technological Development, Impacts, and Careers

Overall expectations

By the end of this course, students will:

B1. Fundamentals of Technological Development

demonstrate an understanding of how various needs and underlying social, economic, and environmental factors drive the evolution of technology

Specific expectations

By the end of this course, students will:

B1.1 assess interrelationships between user needs and the development of various technological solutions

B1.2 analyze how the development and application of technologies are impacted by legal, ethical, social, economic, and environmental considerations

B1.3 investigate and describe contributions to technological innovations made by Canadians, including women, and members of diverse groups and communities in Canada, including First Nations, Métis, and Inuit

B1.4 describe ways in which diverse communities, including First Nations, Métis, and Inuit, have drawn on various knowledge systems to find innovative approaches to technological problem solving

B2. Impacts of Technology

analyze impacts of various technologies on individuals, society, the economy, and the environment

Specific expectations

By the end of this course, students will:

B2.1 assess short-term and long-term impacts of various technological innovations on individuals and society, including the impact on their everyday life, and propose ways to mitigate negative impacts and enhance positive ones

B2.2 assess local and global impacts of various technological innovations on the environment and the economy, including the labour market

B2.3 evaluate how positive and negative impacts of various technologies can influence technological evolution, including emerging technologies

B3. Careers and Pathways in Technology and the Skilled Trades

explore and describe careers in technological fields and the skilled trades, and pathways for entering them

Specific expectations

By the end of this course, students will:

B3.1 explore a variety of roles, responsibilities, and opportunities related to current and emerging careers in technological fields, including a variety of broad-based technology areas, and the skilled trades

B3.2 research and identify programs, including in-school job skills programs and community-based programs, related to pathways and careers in technological fields and the skilled trades

B3.3 compare a variety of pathways leading to careers in technological fields and the skilled trades, including their structure and the educational and financial requirements for them

B3.4 evaluate the transferable skills they are developing, identifying areas of strength and growth, and analyze how these skills relate to current and emerging careers in technological fields and the skilled trades

Information for parents

[A parent's guide to Technology and the Skilled Trades, Grades 9 and 10 \(2024\)](#)

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Resources

[Broad-Based Technology Subject Areas – Concepts and Topics, Skills and Applications, and Careers Key Changes – Technology and the Skilled Trades \(TAS1O and TAS2O\)](#)

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